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MODIFIED O-METHYLTRANSFERASES USEFUL FOR PRODUCTION OF O-METHYLATED PHENOLIC NATURAL PRODUCTS

Abstract

Disclosed are modified Type 1 O-methyltransferases (OMTs) and methods of using such modified OMTs. In some forms of the modified OMTs, the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid. In some forms of the modified OMTs, the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

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Background/Summary

REFERENCE TO SEQUENCE LISTING

[0001] The Sequence Listing XML submitted as a file named “UHK_01454_US_ST26.xml,” created on Mar. 6, 2024, and having a size of 109,775 bytes is hereby incorporated by reference pursuant to 37 C.F.R. § 1.834(c)(1).

FIELD OF THE INVENTION

[0002] The disclosed invention is generally in the field of O-methylation of natural products and specifically in the area of modified O-methyltransferases that methylate stilbenes, and related phenylpropanoids.

BACKGROUND OF THE INVENTION

[0003] Stilbenes are specialized metabolites sporadically distributed in the plant kingdom. Grape (*Vitis vinifera*, dicot), peanut (*Arachis hypogaea*, dicot), Scots pine (*Pinus sylvestris*, gymnosperm), *Sorghum* (*Sorghum bicolor*, monocot) and sugarcane (*Saccharum* spp., monocot) are iconic species producing various stilbene compounds upon (a)biotic challenges.sup.1-5. They serve to maintain reactive oxygen species homeostasis and/or protect against microbial attacks.sup.3-6-8. Recently, stilbenes have emerged as prominent candidates for nutritional and pharmaceutical studies due to their anti-aging, anti-neurodegeneration, anti-diabetes and chemo-prevention properties.sup.9-11. Over 200 stilbene-related clinical trials (internet site clinicaltrials.gov/) have been launched and stilbenes like resveratrol and pterostilbene are commercialized as dietary supplements. Despite the heightened interest in this minor, yet unique, class of phenylpropanoids as pharmaceuticals and nutraceuticals, in planta stilbene biosynthesis remains understudied, especially the downstream derivatization steps such as O-methylations and prenylations. Elucidation of the underlying enzymology would expedite bioengineering attempts to fortify crops with more or novel stilbenes.

[0004] O-Methyltransferases (OMTs) are indispensable in a diversity of metabolic pathways and hence biophysiological processes. For instance, *Sorghum* SbOMT3 catalyzes the biosynthesis of sorgoleones which are allelochemicals exuded by root hairs to suppress competing weeds.sup.12. Two maize OMTs regioselectively methylate flavonoid phytoalexins which inhibit *Fusarium* spp. infection.sup.13. The degree of O-methylations of anthocyanidin pigments affects floral and fruit colors in flowering plants.sup.14-15. In *Arabidopsis* (*Arabidopsis thaliana*), disruption of a tapetum-specific OMT (AtTSM1) which methylates polyamine conjugates adversely affects silique development and seed setting.sup.16. Additionally, the degree of phenylpropanoid O-methylation determines lignin composition in tracheophytes.sup.17. For example, disruption of caffeic acid O-methyltransferase (COMT) and/or caffeoyl-CoA O-methyltransferase (CCoAOMT) substantially alters lignin content and composition, potentially affecting plant growth and development.sup.18-20. Biochemical and structural characterization of these phenylpropanoid OMTs have provided insights into their substrate enzyme interactions and catalytic mechanisms.sup.21-24, hence facilitating genetic manipulations of lignin composition in biomass crop.sup.25, 26.

[0005] Phylogenetically-unrelated plant species have independently recruited specific OMTs for stilbene O-methylation. In response to environmental stresses, grapevine resveratrol OMT (VvROMT) methylates resveratrol to pterostilbene whereas Scots pine pinosylvin OMT (PsPMT2) methylates pinosylvin to pinosylvin monomethyl ether.sup.27, 28. In addition, a root hair-specific sorgoleone O-methyltransferase from *Sorghum*, SbOMT3, methylates resveratrol to pterostilbene in vitro or in transgenic models, albeit at poor efficiency.sup.12, 29. Synthetic biology approaches have been initiated to generate O-methylated stilbenes of pharmaceutical significance such as pterostilbene and 3'-hydroxypterostilbene.sup.30, 31, which are critically acclaimed for their potent

cancer-chemopreventive.sup.32 and neuroprotective properties.sup.33. In fact, their pharmacological properties are superior to the non-O-methylated analogs.sup.3435. However, the lack of stilbene O-methyltransferase (SOMT) structural data greatly limits our understanding of their distinctive substrate specificities, regioselectivities and catalytic efficiencies, impeding bioengineering efforts in the production of high-value and/or novel O-methylated stilbenes. [0006] Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each claim of this application.

BRIEF SUMMARY OF THE INVENTION

[0007] Disclosed are modified Type 1 O-methyltransferases (OMTs) and methods of using such modified OMTs. In some forms of the modified OMTs, the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid. In some forms of the modified OMTs, the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

[0008] In some forms, the substituting polar amino acid is independently Asn, Gln, Thr, or Ser. In some forms, the substituting polar amino acid is Asn. In some forms, the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala. In some forms, the substituting hydrophobic amino acid is Phe.

[0009] In some forms, the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCAldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCAldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SiCOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIEOMT, *Planctopirus limnophila* PlIOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeuruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

[0010] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93. In some forms, the modified OMT has at least 70% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0011] In some forms, the modified OMT has at least 75% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0012] In some forms, the modified OMT has at least 80% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0013] In some forms, the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0014] In some forms, the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0015] In some forms, the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0016] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

[0017] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

[0018] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

[0019] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

[0020] In some forms, the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

[0021] Also disclosed are solutions including a modified OMT as disclosed. In some forms, the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions. Also disclosed are plants that express a modified OMT as disclosed.

[0022] Also disclosed are methods of producing an O-methylated stilbenoid or hydroxycinnamic acid. In some forms, the method involves bringing into contact, in a solution, a modified OMT as disclosed, a substrate for the modified OMT, and S-adenosylmethionine. In general, the O-methylated stilbenoid or hydroxycinnamic acid is produced by incubating the solution under suitable conditions.

[0023] In some forms, the method involves growing that express a modified OMT as disclosed and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

[0024] Additional advantages of the disclosed method and compositions will be set forth in part in the description which follows, and in part will be understood from the description, or can be learned by practice of the disclosed method and compositions. The advantages of the disclosed method and compositions will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIGS. 1A-1F are schematic and plots showing biosynthesis of stilbenes and flavonoids in *Sorghum* and stilbene profiles of *Colletotrichum sublineola*-infected *Sorghum* mesocotyls. FIG. 1A illustrate biosynthetic pathways in stilbene and flavonoid in *Sorghum*. Newly formed methoxy

groups are enlarged. STS, stilbene synthase; SOMT, stilbene O-methyltransferase; COMT: caffeic acid O-methyltransferase; CHS, chalcone synthase; CHI; chalcone isomerase; FNR, flavanone 4-reductase; DFR, dihydroflavonol 4-reductase; FNSII, flavone synthase II; Sb, *Sorghum bicolor*. R=H, OH or OCH₃. FIG. 1B is a plot showing distribution of stilbene aglycones in *C. sublineola*-infected mesocotyls of *Sorghum* genotypes SC748-5 (resistant) and BTx623 (susceptible) 96 h after treatment. Values refer to means±SD (n=3). FIGS. 1C-1F are plots showing HPLC-QTOF-HRMS detection of resveratrol (FIG. 1C), piceid (FIG. 1D), pinostilbene (FIG. 1E) and pterostilbene (FIG. 1F) in *C. sublineola*-infected mesocotyls of *Sorghum* genotypes SC748-5 (resistant) and BTx623 (susceptible). inf, infected; ctrl, control; cps, counts per second.

[0026] FIGS. 2A-2D are plots showing catalytic activities and phylogeny of SbSOMT and SbCOMT. FIG. 2A is a pair of plot showing formation of pinostilbene and pterostilbene from resveratrol by SbSOMT and SbCOMT after two-hour incubation with 100 μM resveratrol. Values refer to means±SD (n=3). n.d.: not detected. Dots represent individual data points. FIG. 2B is plot showing formation of ferulic acid from caffeic acid by SbSOMT and SbCOMT after two-hour incubation with 100 μM phenylpropanoid substrates. Values refer to means±SD (n=3). n.d.: not detected. Dots represent individual data points. FIG. 2C is a plot showing formation of sinapic acid from 5-hydroxyferulic acid by SbSOMT and SbCOMT after two-hour incubation. Values refer to means±SD (n=3). n.d.: not detected. FIG. 2D demonstrates phylogenetic analysis of SbSOMT and SbCOMT. The unrooted phylogenetic tree was constructed by maximum likelihood using MEGA X^{sup.90}. Bootstrapping with 1,000 replications was carried out. Bona fide stilbene O-methyltransferases from gymnosperm^{sup.27}, dicot^{sup.28} and monocot (this study) are highlighted. *Sorghum* OMTs are bolded. Scale bar denotes 0.2 amino acid substitution per site.

[0027] FIGS. 3A-3E are plots showing genotypes, phenotypes and metabolite analysis of *Sorghum* sbsomt mutants. FIG. 3A is a schematic showing genotypes of *Sorghum* sbsomt mutants. Nucleotides corresponding to the protospacer adjacent motif (PAM) are bolded. CRISPR/Cas9-mediated mutations in sbsomt mutant lines are underlined. Sequences are SEQ ID NO:94 (WT, exon 1), SEQ ID NO:95 (WT, exon 2), SEQ ID NO:96 (sbsomt-a, exon 1), SEQ ID NO:97 (sbsomt-a, exon 2), SEQ ID NO:98 (sbsomt-b1, exon 1), SEQ ID NO:99 (sbsomt-b1, exon 2), SEQ ID NO:100 (sbsomt-b2, exon 1), and SEQ ID NO:101 (sbsomt-b2, exon 2). FIGS. 3B-3E are plots showing HPLC-QTOF-HRMS detection of resveratrol (FIG. 3B), piceid (FIG. 3C), pinostilbene (FIG. 3D) and pterostilbene (FIG. 3E) in *C. sublineola*-infected mesocotyls of *Sorghum* sbsomt mutants after 96 h. inf, infected; ctrl, control; cps, counts per second.

[0028] FIGS. 4A-4F demonstrate regioselective O-methylation of piceatannol and three-dimensional structure of SbSOMT and SbCOMT. FIG. 4A show structural comparison between resveratrol, caffeic acid, and piceatannol. Structural similarities are colored accordingly (pink: resveratrol-piceatannol; blue: caffeic acid-piceatannol). FIG. 4B show schematic of regioselective O-methylation of piceatannol catalyzed by SbSOMT (in pink) and SbCOMT (in blue). FIG. 4C Formation of different A- and B-ring O-methylated products from piceatannol by SbSOMT and SbCOMT after two-hour reaction with 100 μM stilbenes. Values refer to means±SD (n=3). n.d.: not detected. Dots represent individual data points. FIG. 4D represents a global structure of dimeric SbSOMT-resveratrol-β-NAD ternary complex solved at 1.72 Å resolution. In which, the two protomers are colored in white and grey, respectively. FIG. 4E highlights the functional domains, namely dimerization domain, substrate binding pocket and SAM-binding domain of a SbSOMT protomer and position of co-crystallized resveratrol (in black) bound to the substrate binding pocket of SbSOMT. FIG. 4F superimposed SbSOMT (in white) and SbCOMT (in grey; retrieved from PDB: 4PGH^{sup.21}) to show the high structural similarity between SbSOMT and SbCOMT (RMSD=2.322 Å) and the elongated threonine-rich loop (Val103-Cys114) unique to SbSOMT.

[0029] FIGS. 5A-5F are structural models showing interactions between stilbenes and SbSOMT catalytic residues, and docking positions of stilbenes with SbSOMT and SbCOMT. FIG. 5A represents a close-up view of SbSOMT substrate binding pocket revealed resveratrol conformation

and resveratrol-interacting residues and water. Dashed line indicates hydrogen bond formation with the distance labelled. FIGS. 5B-5D are structural models that illustrate close-up view on interactions between SbSOMT catalytic residues (His282, Asp283, Glu310 and Glu342) and resveratrol (FIG. 5B), pinostilbene (FIG. 5C), and pterostilbene (FIG. 5D). Dashed line indicates hydrogen bond formation with the distance labelled. FIG. 5E shows the docking conformation of piceatannol ligand (in black) in SbSOMT (left) and SbCOMT (right) along with the surrounding residues-of-interest and calculated binding energy. Dashed line indicates hydrogen bond formation with the distance labelled. FIG. 5F shows the two docking conformations of resveratrol in SbCOMT along with the surrounding residues-of-interest and calculated binding energy. The carbon backbone of non-productive conformation is coloured in black, and the productive conformation is coloured in grey, respectively. Dashed line indicates hydrogen bond formation with the distance labelled.

[0030] FIGS. 6A-6D are plots demonstrating validation of catalytic residues and key substrate-orientating residues of SbSOMT via site-directed mutagenesis. FIGS. 6A-6B are a pair of plots showing formation of pinostilbene (FIG. 6A) and pterostilbene (FIG. 6B) by catalytic residue mutant proteins over a one-hour incubation period. Data are expressed as yields relative to that of wild-type SbSOMT protein (WT). FIGS. 6C-6D are a pair of plots showing formation of pinostilbene (FIG. 6C) and pterostilbene (FIG. 6D) by key substrate-orientating residue mutant proteins over a one-hour incubation period. Data are expressed as yields relative to that of wild-type SbSOMT protein (WT). Values refer to means $\pm$ SD (n=3). Asterisks indicate significant differences between mutant protein and WT (Student's t-test, ***: p<0.001). Tricin was used as an internal standard for quantitation. Note that pterostilbene represents the major product for SbSOMT(WT)-resveratrol reactions at the 2-hour end point (FIG. 2A). n.d.: not detected. Dots represent individual data points.

[0031] FIGS. 7A-7C illustrate biosynthesis of stilbenes in wild sugarcane, stilbene profiles of mechanically-wounded wild sugarcane stalks, and in vitro enzyme activities of SsCOMT. FIG. 7A is a schematic showing proposed stilbene pathways in wild sugarcane. Newly formed methoxy groups are colored and bolded. STS, stilbene synthase; COMT: caffeic acid O-methyltransferase; Ss, *Saccharum spontaneum*. FIG. 7B is a plot showing distribution of stilbene aglycones in mechanically-wounded wild sugarcane stalks 120 h after treatment. Values refer to means $\pm$ SD (n=3). FIG. 7C is a plot showing formation of ferulic acid, sinapic acid, isorhapontigenin, pinostilbene, and pterostilbene from caffeic acid, 5-hydroxyferulic acid, piceatannol, and resveratrol respectively, by SsCOMT, after two-hour incubation with 100 μ M substrates. Tricin was used as an internal standard for quantitation. Values refer to means $\pm$ SD (n=3).

[0032] FIGS. 8A-8D are plots showing expression analysis of pterostilbene biosynthesis related candidate genes in *Sorghum*. FIG. 8A shows analysis of in silico gene expression. RNA-sequencing datasets derived from *Bipolaris sorghicola*-infected *Sorghum* leaves and *Colletotrichum sublineola*-infected *Sorghum* mesocotyls were obtained from literature.^{sup.40, 41} FIGS. 8B-8D are plots showing quantitative RT-PCR gene expression analysis of SbSTS1 (FIG. 8B), SbOMT4 (FIG. 8C) and SbSOMT (FIG. 8D). Expression levels are expressed relative to *Sorghum bicolor* Eukaryotic Initiation Factor 4A-1 (SbEIF4 α).^{sup.68} Values refer to means $\pm$ SD (n=3). inf, infected; ctrl, control.

[0033] FIGS. 9A-9C demonstrate initial screening of catalytic activities of *Sorghum* OMTs. FIG. 9A represents chemical structures showing in vitro O-methylation of resveratrol by *Sorghum* OMTs. FIGS. 9B-9C are a pair of plots showing HPLC-QTOF-HRMS detection of pinostilbene (FIG. 9B), and pterostilbene (FIG. 9C) in *Sorghum* OMTs catalyzed reaction using resveratrol as a substrate. cps, counts per second; NC, negative control (reaction without enzyme).

[0034] FIGS. 10A-10D demonstrate co-overexpression of SbSTS1 and SbSOMT in *Nicotiana benthamiana* leaves. FIG. 10A is a schematic showing in planta generation of pterostilbene by *Sorghum* SbSTS1 and SbSOMT. FIGS. 10B-10D are plots showing HPLC-QTOF-HRMS detection

of resveratrol (FIG. 10B), pinostilbene (FIG. 10C), and pterostilbene (FIG. 10D) in *Nicotiana benthamiana* leaves transiently expressing SbSTS1 or co-expressing SbSTS1 and SbSOMT. Cps, counts per second.

[0035] FIG. 11A-11F are plots showing genotyping of *Sorghum* sbsomt mutants. FIGS. 11A-11F are representative chromatographs of direct sequencing of the target sites in the sbsomt-a (FIGS. 11A-11B), sbsomt-b1 (FIGS. 11C-11D), and sbsomt-b2 (FIGS. 11E-11F) mutant lines (T.sub.1 generation) are shown. CRISPR/Cas9-mediated mutations in sbsomt mutant lines are underlined. Bold, protospacer adjacent motif (PAM) site. Sequences are SEQ ID NO:94 (reference, target 1), SEQ ID NO:95 (reference, target 2), SEQ ID NO:96 (sbsomt-a, target 1), SEQ ID NO:97 (sbsomt-a, target 2), SEQ ID NO:98 (sbsomt-b1, target 1), SEQ ID NO:99 (sbsomt-b1, target 2), SEQ ID NO:100 (sbsomt-b2, target 1), and SEQ ID NO:101 (sbsomt-b2, target 2).

[0036] FIG. 12 demonstrates predicted effects of mutations in SbSOMT in *Sorghum* sbsomt mutant lines. The amino acid sequences of wild-type and mutant SbSOMT were aligned using Clustal Omega.sup.88. First premature stop codons (*) are underlined and bolded. The catalytic residues of *Sorghum* SbSOMT are underlined and bolded. Amino acid residues contributing to the hydrophobic interactions with stilbene substrates are labelled with t above the corresponding residues. Sequences are SEQ ID NO:1 (WT), SEQ ID NO:102 (sbsomt-a), SEQ ID NO:103 (sbsomt-b1), and SEQ ID NO:104 (sbsomt-b2).

[0037] FIG. 13 demonstrates multiple sequence alignment of bonafide stilbene O-methyltransferases (SOMTs). The amino acid sequence of SbSOMT, VvROMT and PsPMT2 were aligned using Clustal Omega.sup.88. The catalytic residues of *Sorghum* SbSOMT are underlined and bolded, while those of other SOMTs are only underlined. Amino acid residues contributing to the hydrophobic interactions with stilbene substrates are labelled with t above the corresponding residue. Fully conserved amino acid residues across all three SOMTs are indicated by asterisks; those with strongly similar properties are indicated with colons; those with weakly similar properties are indicated with periods. Sequences are SEQ ID NO:1 (SbSOMT), SEQ ID NO:27 (VvROMT), and SEQ ID NO:14 (PsPMT2).

[0038] FIG. 14 is a schematic showing phylogeny of wild sugarcane OMTs. Phylogenetic analysis of wild sugarcane OMTs. The unrooted phylogenetic tree was constructed by maximum likelihood using MEGA X.sup.90. Bootstrapping with 1,000 replications was carried out. Scale bar denotes 0.5 amino acid substitution per site. The sequences of the noted genes are hereby incorporated by reference.

[0039] FIGS. 15A-15X are titration thermograms and fitting curves of conducted ITCs. Each set of thermogram (above) and fitting curve (below) represents performed ITC of different combinations of protein (in cell) and ligand (in syringe) as titled.

[0040] FIGS. 16A-16C are structures obtained in this study. FIG. 16A is superimposed SbSOMT-resveratrol complex with two SbSOMT-stilbene- β -NAD complexes obtained in this study towards SbSOMT-resveratrol- β -NAD (top left). SbSOMT-resveratrol complex (top right; RMSD=0.468 Å), SbSOMT-pinostilbene- β -NAD, (bottom left; RMSD=0.995 Å) and SbSOMT-pterostilbene- β -NAD (bottom right; RMSD=0.730 Å) complexes all shares high structural similarity towards SbSOMT-resveratrol- β -NAD. FIG. 16B is a close-up view focusing the conformations of superimposed stilbene derivatives (in black) and correspondent surrounding residues. Leu28 (in grey) is constituted in adjacent protomer. FIG. 16C illustrate the FO-FC electron density omit maps of resveratrol and β -NAD ligands in SbSOMT-resveratrol- β -NAD ternary complex contoured at 3σ (mesh) and -3σ (blob).

[0041] FIG. 17 is a schematic showing reaction mechanism of consecutive stilbene O-methylations mediated by SbSOMT. Schematic of SbSOMT O-methylation cycle exemplified with S-adenosyl methionine (SAM, donor) and resveratrol (acceptor).sup.21, 22, 43, 44. Upon binding of SAM and resveratrol (Step I), the surrounding electrostatic interactions allow His282 N.sub.ε to deprotonate 3-OH of resveratrol. Step II, deprotonated O.sup.- catalyzes SN.sub.2 attack towards methyl group

of SAM sulfonium ylide to produce pinostilbene and S-adenosyl-homocysteine (SAH). Step III, the His282 protonated imidazole is regenerated by solvent and returns to the neutral state. Step IV, the SAH and pinostilbene are dissociated from SbSOMT.

[0042] FIGS. **18A-18C** are plots showing stilbene profiles of mechanically-wounded wild sugarcane stalks. FIGS. **18A-18C** demonstrate HPLC-QTOF-HRMS detection of resveratrol (FIG. **18A**), piceatannol (FIG. **18B**), and isorhapontigenin (FIG. **18C**) in mechanically-wounded wild sugarcane stalks 120 h after treatment. Cps, counts per second.

[0043] FIGS. **19A-19B** are a pair of plots showing tentative identification of piceatannol-hexoside from mechanically wounded wild sugarcane stalks. FIG. **19A** demonstrates HPLC-QTOF-HRMS detection of putative piceatannol-hexoside. FIG. **19B** demonstrates MS^{sup.2} fragmentation pattern of putative piceatannol-hexoside. A neutral loss of 162 Da, potentially corresponding to an anhydrohexose unit, was observed for this ion. Glc: glucose; cps, counts per second.

[0044] FIGS. **20A-20D** are plots showing expression analysis of stilbene biosynthesis related genes in wild sugarcane. FIGS. **20A-B** are a pair of plots showing quantitative RT-PCR gene expression analysis of SsSTS (FIG. **20A**) and SsCOMT (FIG. **20B**). Expression levels are expressed relative to *Saccharum spontaneum* Glyceraldehyde 3-Phosphate Dehydrogenase (SsGADPH)^{sup.68}. Values refer to means±SD (n=5). FIGS. **20C-20D** are semi-quantitative RT-PCR gene expression analysis of SsSTS (FIG. **20C**) and SsCOMT (FIG. **20D**) relative to SsGADPH. Thirty cycles were used for amplification. Two biological duplicates from each time point were analyzed.

[0045] FIG. **21** represents multiple sequence alignment of SbSTS1 and SsSTS. The amino acid sequences of SbSTS1 and SsSTS were aligned using Clustal Omega^{sup.88}. Fully conserved amino acid residues are indicated by asterisks; those with strongly similar properties are indicated with colons; those with weakly similar properties are indicated with periods. The sequences are SEQ ID NO:23 (SbSTS1) and SEQ ID NO:33 (SsSTS).

[0046] FIGS. **22A-22E** demonstrate catalytic activities of SsSTS. FIG. **22A** is a schematic showing proposed STS reaction with feruloyl-CoA and malonyl-CoA as substrates. FIGS. **22B-22D** are plots showing HPLC-QTOF-HRMS detection of isorhapontigenin (FIG. **22B**), 3'-methoxybisinoryangonin (FIG. **22C**), and feruloyltriacetic acid lactone (FIG. **22D**) in SsSTS catalyzed reaction using feruloyl-CoA and malonyl-CoA as substrates. The detection of these derailment products is consistent with a previous report showing that feruloyl-CoA is a poor substrate for STSs, at least under in vitro conditions^{sup.5}. FIG. **22E** is a plot showing HPLC-QTOF-HRMS detection of piceatannol in SsSTS catalyzed reaction using caffeoyl-CoA and malonyl-CoA as substrates. cps, counts per second; NC, negative control (reaction without enzyme).

[0047] FIG. **23** represents multiple sequence alignment of SbCOMT, SbSOMT, and SsCOMT. The amino acid sequences of SbCOMT, SbSOMT, and SsCOMT were aligned using Clustal Omega^{sup.88}. Fully conserved amino acid residues are indicated by asterisks; those with strongly similar properties are indicated with colons; those with weakly similar properties are indicated with periods. Residues of SbSOMT and SsCOMT corresponding to SbCOMT Asn128 and Asn323 were underlined and bolded. Sequences SEQ ID NO:1 (SbSOMT), SEQ ID NO:2 (SbCOMT), and SEQ ID NO:32 (SsCOMT).

[0048] FIGS. **24A-24I** show catalytic activities of SsCOMT. FIG. **24A** represents in vitro O-methylation of piceatannol by SsCOMT. FIG. **24B** is a plot showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of isorhapontigenin in SsCOMT catalyzed reaction using resveratrol as a substrate. FIG. **24C** represents in vitro O-methylation of caffeic acid by SsCOMT. FIG. **24D** is a plot showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of ferulic acid in SsCOMT catalyzed reaction using caffeic acid as a substrate. FIG. **24E** represents in vitro O-methylation of 5-hydroxyferulic acid by SsCOMT. FIG. **24F** is a plot showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of sinapic acid in SsCOMT catalyzed reaction using 5-hydroxyferulic acid as a substrate.

FIG. 24G represents in vitro O-methylation of resveratrol by SsCOMT. FIGS. 24H-24I are a pair of plots showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of pinostilbene (FIG. 24H), and pterostilbene (FIG. 24I) in SsCOMT catalyzed reaction using resveratrol as a substrate. cps, counts per second; NC, negative control (reaction without enzyme). [0049] FIG. 25 is a structural model that explains the basis of polarity pairing via the representative SbSOMT-resveratrol interaction. The structure shows that at the proximity of catalytic dyad (Hiscat and Aspcat), the non-polar Phe337 of SbSOMT is closely positioned to non-polar C4 (the vicinal group of reactive site of resveratrol) and demonstrate the coordination of substrate orientation via polarity pairing. In SbCOMT context (bottom right box), the polar Asn323 is positioned at the equivalent location of Phe337 of SbSOMT. Hence, SbCOMT coordinates via the opposite polar-to-polar polarity pairing.

[0050] FIGS. 26A-26X are structural models showing polarity pairing as a common regioselective mechanism across OMT-ligand complexes deposited to PDB. Twenty-four unique OMT-ligand complexes were retrieved from PDB (by 10 Aug. 2022), and their respective amino acid residue equivalent to SbCOMT.sup.Asn323/SbSOMT.sup.Phe337, catalytic dyad, and bound ligand were shown. Homolog search was used to retrieve ($\geq 30\%$ identity to SbCOMT or SbSOMT) and screened for complexes that are bound to ligand at substrate binding pocket. Further details are listed in Table 10. Appropriate polarity pairing between this residue's side chain and the functional group vicinal to the reactive site (methyl-accepting —OH group) was observed in all complexes, except for the three structures excluded from the analysis employed herein, either due to the use of non-reactive ligand (3I5U and 5ICF) or non-productive OMT-ligand conformation (5I2H). The carbon vicinal to the reactive site is circled in black.

[0051] FIGS. 27A-27M demonstrate in vitro toxicity of stilbenes towards *Colletotrichum sublineola*. FIGS. 27A-27I are images showing mycelial growth of *C. sublineola* on potato dextrose agar after a 72-hour incubation with 4% (v/v) DMSO (FIG. 27A), 50 μ M resveratrol (FIG. 27B), 50 μ M pinostilbene (FIG. 27C), 5 μ M pterostilbene (FIG. 27D), 10 μ M pterostilbene (FIG. 27E), 20 μ M pterostilbene (FIG. 27F), 30 μ M pterostilbene (FIG. 27G), 40 μ M pterostilbene (FIG. 27H) and 50 μ M pterostilbene (FIG. 27I). Scale bars denote 5 mm. FIGS. 27J-27L are images that show spore germination of *C. sublineola* on potato dextrose agar after a 12-hour incubation in darkness with 4% (v/v) DMSO (FIG. 27J), 40 μ M pterostilbene (FIG. 27K) and 50 μ M pterostilbene (FIG. 27L). Scale bars denote 500 μ m. FIG. 27M is a plot showing diameter of *C. sublineola* colonies after a 72-hour incubation with different concentrations of pterostilbene. IC₅₀ was estimated by GraphPad Prism 6. Values refer to means $\pm$ SD (n=9). IC₅₀, half-maximal inhibitory concentration.

[0052] FIG. 28 is an image showing arrangement of SbSTS1 (Sb07g004700; Sobic.007G058900), SbOMT4 (Sb07g004690; Sobic.007G058800), and SbSOMT (Sb07g004710; Sobic.007G059100) in *Sorghum* genome. Gene arrangement in Chr07:6094677 . . . 6186377 of *Sorghum* genome was obtained from Phytozome v13.sup.65.

[0053] FIGS. 29A-29L are plots showing MSMS spectrum of stilbenes detected in plant samples and authentic standards. FIGS. 29A-29B are representative MSMS spectrum of resveratrol detected by HPLC-QTOF-HRMS in (FIG. 29A) *C. sublineola*-infected *Sorghum* mesocotyls or mechanically-wounded wild sugarcane, and (FIG. 29B) authentic resveratrol standard. FIGS. 29C-29D are representative MSMS spectrum of piceid detected by HPLC-QTOF-HRMS in (FIG. 29C) *C. sublineola*-infected *Sorghum* mesocotyls, and (FIG. 29D) authentic piceid standard. FIGS. 29E-29F are representative MSMS spectrum of pinostilbene detected by HPLC-QTOF-HRMS in (FIG. 29E) *C. sublineola*-infected *Sorghum* mesocotyls, and (FIG. 29F) authentic pinostilbene standard. FIGS. 29G-29H are representative MSMS spectrum of pterostilbene detected by HPLC-QTOF-HRMS in (FIG. 29G) *C. sublineola*-infected *Sorghum* mesocotyls, and (FIG. 29H) authentic pterostilbene standard. FIGS. 29I-29J are representative MSMS spectrum of piceatannol detected by HPLC-QTOF-HRMS in (FIG. 29I) mechanically-wounded wild sugarcane, and (FIG. 29J)

authentic piceatannol standard. FIGS. 29K-29L are representative MSMS spectrum of isorhapontigenin detected by HPLC-QTOF-HRMS in (FIG. 29K) mechanically-wounded wild sugarcane, and (FIG. 29L) authentic isorhapontigenin standard. cps, counts per second; m/z, mass-to-charge ratio.

DETAILED DESCRIPTION OF THE INVENTION

[0054] The disclosed method and compositions can be understood more readily by reference to the following detailed description of particular embodiments and the Example included therein and to the Figures and their previous and following description.

[0055] O-methyltransferases (OMTs) are ubiquitous enzymes in plants and are involved in a type of chemical modification, known as O-methylation, of phenolic natural products. Methylation of these compounds generally enhances their hydrophobicity, which confers higher bioavailability after intake and thus significantly increases the potency of their health-promoting properties. A biochemical mechanism termed compatible polarity pairing has been discovered that determines the catalytic regioselectivity in O-methylation of phenolic hydroxyl (—OH) groups in Type 1 OMTs. The discovery is based on the identification of a key amino acid position within Type 1 OMTs (corresponding or equivalent to Phe337 of SbSOMT), which, if the said amino acid is highly polar (or hydrophilic, e.g. asparagine) in nature, preferentially interacts with the polar functional group (e.g. —OH and —OMe) of an incoming aromatic ring, thus positioning the —OH group(s) adjacent to the polar functional group for O-methylation. In contrast, if the said amino acid is hydrophobic in nature (e.g. phenylalanine, valine, threonine), it preferentially interacts with an aromatic carbon in the phenolic ring of an incoming substrate, thus positioning the —OH group(s) adjacent to the aforementioned aromatic carbon for O-methylation. Thus, a structure-function relationship for the nature of the amino acid that corresponds to Phe337 of SbSOMT allows prediction and alteration of the regioselectivity of substrated for modified OMTs.

[0056] In the case of SbSOMT, Phe337 interacts with C4 of the stilbene A-ring and the C3-OH group is positioned for O-methylation. On the other hand, the equivalent residue in SbCOMT is Asn323 which interacts with 4'-OH group of the stilbene B-ring, hence t3'-OH group is positioned for O-methylation. Consistent with this, we discovered a panel of 24 Type 1 OMT-ligand complexes that supports this same compatible polarity pairing between the amino acid residue equivalent to SbSOMT Phe337 and SbCOMT Asn323 and their substrates. These Type 1 OMT enzymes all utilize the same catalytic mechanisms to catalyze O-methylation of phenolic substrates. Hence, our discovery allows the predictable engineering of Type 1 OMTs with an appropriate residue at the corresponding position for regiospecific O-methylation of phenolic substrates.

[0057] Previous attempts at biosynthesis of O-methylated phenolic compounds relied on the trial-and-error approach. They relied solely on the natural capability of Type 1 OMTs to catalyze O-methylation of phenolic substrates in vivo. Hence, both the OMTs involved and the resultant end products were unpredictable and had to be experimentally tested which is time- and resource-consuming. Optimization of Type 1 OMT-regioselectivity to tailor make the end products was very challenging and, again, relied on trial-and-error. Thus, the mechanism underlying the catalytic regioselectivity of O-methyltransferases at the heart of our discovery had long been sought after.

[0058] With our discovery, biosynthesizing O-methylated phenolic natural products in a biosystem now becomes highly precise and predictable. Our discovery also allows the synthesis of novel O-methylated phenolic compounds.

[0059] Our discovery allows (1) straightforward prediction of Type 1 OMT(s) responsible for catalyzing certain —OH groups of phenolic substrates, (2) rational choice of Type 1 OMT(s) for targeted compound biosynthesis and synthetic biology, and (3) optimization of Type 1 OMT catalytic activity and regioselectivity by protein engineering.

[0060] The underlying biochemical mechanism determining the regioselectivity of Type 1 OMTs was largely unknown until our discovery here. Accordingly, the incorporation of a hydrophobic

amino acid at position equivalent to SbSOMT Phe337 in any OMT will greatly favor the O-methylation of —OH group(s) adjacent to the interacting aromatic carbon of a phenolic ring in a substrate. In contrast, the incorporation of a hydrophilic amino acid at position equivalent to SbSOMT Phe337 in any OMT will greatly favor the O-methylation of —OH group(s) adjacent to an interacting polar functional group of a phenolic ring in a substrate.

[0061] Disclosed are modified Type 1 O-methyltransferases (OMTs) and methods of using such modified OMTs. In some forms of the modified OMTs, the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid. In some forms of the modified OMTs, the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

[0062] In some forms, the substituting polar amino acid is independently Asn, Gln, Thr, or Ser. In some forms, the substituting polar amino acid is Asn. In some forms, the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala. In some forms, the substituting hydrophobic amino acid is Phe.

[0063] In some forms, the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCaldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCaldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SiCOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIEOMT, *Planctopirus limnophila* PlIOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

[0064] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93. In some forms, the modified OMT has at least 70% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0065] In some forms, the modified OMT has at least 75% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0066] In some forms, the modified OMT has at least 80% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0067] In some forms, the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0068] In some forms, the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0069] In some forms, the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34,

35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0070] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

[0071] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

[0072] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

[0073] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

[0074] In some forms, the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

[0075] Also disclosed are solutions including a modified OMT as disclosed. In some forms, the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions. Also disclosed are plants that express a modified OMT as disclosed.

[0076] Also disclosed are methods of producing an O-methylated stilbenoid or hydroxycinnamic acid. In some forms, the method involves bringing into contact, in a solution, a modified OMT as disclosed, a substrate for the modified OMT, and S-adenosylmethionine. In general, the O-methylated stilbenoid or hydroxycinnamic acid is produced by incubating the solution under suitable conditions.

[0077] In some forms, the method involves growing that express a modified OMT as disclosed and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

[0078] It is to be understood that the disclosed method and compositions are not limited to specific synthetic methods, specific analytical techniques, or to particular reagents unless otherwise specified, and, as such, can vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

[0079] The disclosed compositions and methods can be further understood through the following numbered paragraphs.

[0080] 1. A modified Type 1 O-methyltransferase (OMT), wherein the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid or wherein the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid.

[0081] 2. The modified OMT of paragraph 1, wherein the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid or wherein the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

[0082] 3. The modified OMT of paragraph 1 or 2, wherein the substituting polar amino acid is independently Asn, Gln, Thr, or Ser.

[0083] 4. The modified OMT of any one of paragraphs 1-3, wherein the substituting polar amino acid is Asn.

[0084] 5. The modified OMT of any one of paragraphs 1-4, wherein the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala.

[0085] 6. The modified OMT of any one of paragraphs 1-5, wherein the substituting hydrophobic

amino acid is Phe.

[0086] 7. The modified OMT of any one of paragraphs 1-6, wherein the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCAldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCAldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SICOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIEOMT, *Planctopirus limnophila* PlIOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

[0087] 8. The modified OMT of any one of paragraphs 1-7, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0088] 9. The modified OMT of any one of paragraphs 1-8, wherein the modified OMT has at least 70% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0089] 10. The modified OMT of any one of paragraphs 1-9, wherein the modified OMT has at least 75% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0090] 11. The modified OMT of any one of paragraphs 1-10, wherein the modified OMT has at least 80% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0091] 12. The modified OMT of any one of paragraphs 1-11, wherein the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0092] 13. The modified OMT of any one of paragraphs 1-12, wherein the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0093] 14. The modified OMT of any one of paragraphs 1-13, wherein the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0094] 15. The modified OMT of any one of paragraphs 1-14, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

[0095] 16. The modified OMT of any one of paragraphs 1-15, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted

with a hydrophobic amino acid.

[0096] 17. The modified OMT of any one of paragraphs 1-16, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

[0097] 18. The modified OMT of any one of paragraphs 1-17, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

[0098] 19. The modified OMT of any one of paragraphs 1-18, wherein the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

[0099] 20. A solution comprising the modified OMT of any one of paragraphs 1-19, wherein the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions.

[0100] 21. A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising bringing into contact, in a solution, the modified OMT of any one of paragraphs 1-19, a substrate for the modified OMT and S-adenosylmethionine; and incubating under suitable conditions to produce the O-methylated stilbenoid or hydroxycinnamic acid.

[0101] 22. A plant that expresses the modified OMT of any one of paragraphs 1-19.

[0102] 23. A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising growing the plant of paragraph 22 and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

EXAMPLES

[0103] O-Methylated stilbenes are prominent nutraceuticals but rarely produced by crops. Here, the inherent ability of two Saccharinae grasses to produce regioselectively O-methylated stilbenes is reported. A stilbene O-methyltransferase, SbSOMT, is first shown to be indispensable for pathogeninducible pterostilbene (3,5-bis-O-methylated) biosynthesis in *Sorghum* (*Sorghum bicolor*). Phylogenetic analysis indicates the recruitment of genus-specific SOMTs from canonical caffeic acid O-methyltransferases (COMTs) after the divergence of *Sorghum* spp. from *Saccharum* spp. In recombinant enzyme assays, SbSOMT and COMTs regioselectively catalyze O-methylation of stilbene A-ring and B-ring respectively. Subsequently, SOMT-stilbene crystal structures are presented. Whilst SbSOMT shows global structural resemblance to SbCOMT, molecular characterizations illustrate two hydrophobic residues (Ile144/Phe337) crucial for substrate binding orientation leading to 3,5-bis-O-methylations in the A-ring. In contrast, the equivalent residues (Asn128/Asn323) in SbCOMT facilitate an opposite orientation that favors 3'-O-methylation in the B-ring. Consistently, a highly-conserved COMT is likely involved in isorhapontigenin (3'-O-methylated) formation in wounded wild sugarcane (*Saccharum spontaneum*). Altogether, our work reveals the potential of Saccharinae grasses as a source of O-methylated stilbenes, and rationalize the regioselectivity of SOMT activities for bioengineering of O-methylated stilbenes.

[0104] Within the family Poaceae, the *Sorghum* and *Saccharum* genera evolved from a common ancestral grass and belong to the tribe Andropogonae, subtribe Saccharinae. Species from both genera accumulate hydroxylated stilbenes (e.g. resveratrol, piceatannol) and their O-glycosides. sup.5, 7, 36, but O-methylated stilbenes have not been reported in these species so far. Here, we report the identification of O-methylated stilbenes, pinostilbene (3-O-methylated) and pterostilbene (3,5-bis-O-methylated) in *Sorghum*, and isorhapontigenin (3'-O-methylated) in wild sugarcane (*S. spontaneum*). A stilbene OMT (SbSOMT) catalyzes sequential 3,5-bis-O-methylation of resveratrol to pterostilbene whereas a canonical COMT (SsCOMT) likely catalyzes 3'-O-methylation of piceatannol into isorhapontigenin in wild sugarcane. We generated *Sorghum* SbSOMT CRISPR/Cas9 mutants which were deficient in pinostilbene and pterostilbene, hence

establishing the indispensable role of SbSOMT in resveratrol 3,5-bis-O-methylation. As indicated by phylogenetic analysis, divergence of *Sorghum* genus from other genera in the Saccharinae subtribe likely predated the recruitment of SbSOMT from a canonical COMT. By solving the high-resolution crystal structures of SbSOMT complexed with stilbenes, molecular features that differentiate the substrate binding mode and catalytic regioselectivity between SbSOMT and canonical COMTs were unveiled.

Methods

Plant Materials and Fungal Treatment Conditions

[0105] Two *Sorghum* (*Sorghum bicolor*) genotypes, BTx623 and SC748-5, were analyzed in this study. *Sorghum* seeds were germinated in aerated double-deionized water at 30° C. for 24 h. Wild sugarcane (*Saccharum spontaneum*) sets were obtained from Taiwan^{sup.66}, and planted in soil until flowering. *C. sublineola* isolate TX430BB was propagated on 5% (w/v) oatmeal agar for 14 days prior to fungal infection experiments. Etiolated *Sorghum* seedlings were prepared as described previously^{sup.67}, and were either sprayed with a 0.2% (w/v) bovine gelatin solution (control) or a conidial suspension (1.0×10^6 spores ml⁻¹) of *C. sublineola* isolate TX430BB in the same gelatin solution (treatment). Both control and treatment groups were incubated under constant light at room temperature and 100% humidity.

Plant Metabolite Extraction

[0106] One hundred milligrams of *Sorghum* mesocotyls from fungal infection experiments or mechanically wounded wild sugarcane stalks were harvested for metabolite analysis. All plant tissues were frozen by liquid nitrogen and ground into fine powder using TissueLyser II (QIAGEN, Germany). Two hundred microliters of 80% (v/v) methanol (with 10 μ M Apigenin-d_{sub.5} as internal standard) were added to the samples. Metabolites were extracted by ultra-sonicating the samples on an ice-water bath for 30 min. Samples were passed through a 0.22 μ m PTFE membrane filter (Phenomenex, USA) prior to HPLC-QTOF-HRMS analysis described below.

Hplc-QTOF-HRMS and HPLC-MRM Analyses

[0107] To screen for OMT activities, purified stilbene products of enzyme assays were separated by a Synergi C18 column (Synergi 4 μ Fusion RP 80 Å, 50×2 mm, Phenomenex, USA) under a flow rate of 0.5 ml min⁻¹ over a 6 min linear gradient of 10%-90% B (described below). Product detection was achieved with a quadrupole-time-of-flight-high resolution mass spectrometer (QTOF-HRMS) X500R system (AB Sciex, China) operating under the information-dependent acquisition (IDA) mode. Meanwhile, purified phenylpropanoid products of enzyme assays were separated on the same column connected to a HP1100 series HPLC system (Agilent Technologies, USA) linked to an AP3200-QTRAP mass spectrometer (AB Sciex, China) operating under the multiple-reaction-monitoring (MRM) mode. To quantify these products, assay products were separated on a Kinetex C18 column (Kinetex 2.6 μ m C18 100 Å, 100×2.1 mm, Phenomenex, USA) connected to a HP1100 series HPLC system (Agilent Technologies, USA) linked to an AP3200-QTRAP mass spectrometer (AB Sciex, China) operating under the multiple-reaction-monitoring (MRM) mode. A linear gradient of 10%-90% B under a flow rate of 0.3 ml min⁻¹ over 6 min was used to further enhance sensitivity. Total metabolites from all plant tissues were separated by the same Kinetex C18 column connected to the X500R QTOF-HRMS system. A linear gradient under a flow rate of 0.3 ml min⁻¹ over a 20 min linear gradient of 10%-90% B was used.

[0108] In all HPLC-MS analyses, the mobile phase consisted of 0.5% (v/v) formic acid/water (A) and 0.5% (v/v) formic acid/methanol (B). Stilbenes were detected using the positive ionization mode while phenolic acids were detected with the negative ionization mode. Compounds were quantified by integration of peak area using the quantification mode of SCIEX OS (for X500R) or Analyst software version 1.5.6 (for AP3200-QTRAP). Identification of compounds was achieved by comparing the retention time and MS/MS spectra (FIGS. 29A-29L) with authentic standards.

Gene Expression Analyses

[0109] Total RNA was extracted from *Sorghum* mesocotyls collected at a 24-hour interval for up to

96 h post fungal-treatment and wounded wild sugarcane stalks collected at different time points using the TRIzol method (Invitrogen, USA). Reverse transcription and qRT-PCR were performed using PrimeScript RT reagent kit with gDNA eraser (TaKaRa, Japan) and TB Green premix Ex Taq II kit (TaKaRa, Japan), respectively. Semi qRT-PCR was performed with GoTaq DNA polymerase (Promega, USA). Gene-specific primers used for qRT-PCR experiments were listed in Table 11. The housekeeping gene *Sorghum bicolor* Eukaryotic Initiation Factor 4A-1 (SbEIF4 α .sup.68; XM_002451491) and *Saccharum spontaneum* Glyceraldehyde 3-Phosphate Dehydrogenase (SsGADPH; Sspon.08G0001560-1A).sup.69 were used as internal controls for *Sorghum* and wild sugarcane, respectively.

Cloning of SbSTS1, SbOMT4, SbSOMT, SbCOMT, SsSTS, and SsCOMT

[0110] The coding sequences (CDS) of SbSTS1, SbOMT4, SbSOMT and SbCOMT were amplified from cDNA prepared from fungal-infected mesocotyls of *Sorghum* genotype BTx623 using gene-specific primers (Table 11). The CDS of SsSTS and SsCOMT were cloned from cDNA prepared from wounded wild sugarcane stalk using gene-specific primers (Table 11). To generate recombinant proteins, the CDS of SsSTS (full length), SsCOMT (full length), SbCOMT (encoding residues 2-362), and SbSOMT (encoding residues 2-377) were cloned into pET-N-His-TEV (Beyotime) via HiFi DNA Assembly (New England BioLabs, USA) or Gibson Assembly (New England BioLabs, USA). Full length CDS of SbOMT4 was inserted between BamHI and HindIII restriction sites of pET23a(+) vector (Novagen, Germany). The full length CDSs of SbSTS1 and SbSOMT were individually subcloned into the binary vector pEAQ-HT by the Gibson Assembly method (New England BioLabs, USA) using gene-specific primers (Table 11).

Recombinant Protein Production and Site-Directed Mutagenesis

[0111] Desired mutations of SbSOMT were introduced into its construct using specific primers (Table 11). Expression constructs were transformed into Rosetta™ (DE3) competent *Escherichia coli* cells (Novagen, Germany). Protein expression was induced with 0.1 mM Isopropyl- β -thiogalactopyranoside (IPTG) overnight at 16° C. *E. coli* cells were harvested, resuspended in lysis buffer (20 mM Tris buffer pH 7.9, 300 mM NaCl, 2.5 mM β -mercaptoethanol), and lysed by sonication. Crude proteins were loaded onto a HisTrap HP column (Cytiva, USA) connected to a AKTA pure chromatography system (Cytiva, USA). Column was washed with 15 column volumes of lysis buffer with 50 mM imidazole. Proteins were eluted over a gradient of 50-500 mM imidazole. Eluate fractions with the target OMT were pooled, digested with TEV protease, and simultaneously dialyzed against a low salt buffer (20 mM Tris pH 7.9, 50 mM NaCl, 2.5 mM β -mercaptoethanol) overnight at 4° C. The dialyzed sample was filtered and flowed through HisTrap to remove His-tagged components. The flowthrough was then loaded to HiTrap Q FF columns (Cytiva, USA) over a gradient of 50-1000 mM NaCl. Polishing was achieved via size-exclusion chromatography using the storage buffer (100 mM HEPES pH 7.9, 100 mM NaCl) as mobile phase. SsSTS was purified under the same conditions except that the pH was adjusted to 7.0 for all buffers. Proteins were concentrated by ultrafiltration and flash frozen for storage at -80° C.

Enzyme Assays and Enzyme Kinetics

[0112] For initial screening of OMT activities, purified OMT enzymes (10 μ g) were incubated in 100 mM HEPES buffer (pH 7.9), 200 μ M S-adenosyl-L-methionine (SAM) and 100 μ M of phenylpropanoid or stilbene substrates (final volume 200 μ l) at 30° C. for 2 h. Enzyme kinetics were determined by incubating 1 μ g protein with 100 mM HEPES buffer (pH 7.9), 300 μ M SAM, and stilbene concentrations ranging from 5 to 200 μ M at 30° C. for 5 min. Similar conditions were used to determine enzyme kinetics of SbSOMT towards SAM, except that SAM concentrations ranging from 5 to 500 μ M (while stilbene substrates were kept at 1 mM) were included to accommodate its unexpectedly high K_d and K_m values. For the SsSTS enzyme assays, 10 μ g of purified SsSTS enzymes were incubated in 100 mM HEPES buffer (pH 7.0), 100 μ M of caffeoyl-CoA or feruloyl-CoA, and 300 μ M of malonyl-CoA at 30° C. for 1 h. All enzyme assay reactions were quenched by snap-freezing the reaction tubes in liquid nitrogen. Reaction products were

extracted twice with ethyl acetate. The organic layers were then pooled, vacuum-dried, and resuspended in 50 μ l 80% (v/v) methanol with 10 μ M triclinic as internal standard for HPLC-MRM quantification. All reactions were done in triplicates. Enzyme kinetics were calculated using the non-linear regression fitting by GraphPad Prism 6 software (GraphPad, USA).

Isothermal Titration Calorimetry (ITC)

[0113] Isothermal titration calorimetry (ITC) assays were conducted with MicroCal iTC200 (Malvern Panalytical, UK). Concentrated stocks of resveratrol, pinostilbene, pterostilbene, and piceatannol were prepared in 50% (v/v) PEG 400. Both cell sample and titrant were equilibrated to ITC buffer consisting of 100 mM HEPES pH 7.9 with 2.5% (v/v) PEG 400. For ITC assays involving pterostilbene, PEG 400 concentration was increased to 5% (v/v). Concentrations of protein (in cell) and ligand (in syringe) used in each run are listed in Table 11. Each run incorporated an initial delay of 60 s prior to the first injection (0.5 μ l) and spaced 150 s between the subsequent 19 injections (2.0 μ l) at 25° C. During the runs, the cell was continuously stirred at 750 rpm by flat paddle. Results were analysed based on the 2nd to 20th injections using the Origin (MicroCal Software, USA) and PEAQ-ITC software (Malvern Panalytical, UK).

Agroinfiltration of *N. benthamiana* Leaves

[0114] pEAQ constructs harboring either SbSTS1 or SbSOMT were transformed into *Agrobacterium tumefaciens* strain GV3101. To co-express SbSTS1 and SbSOMT in *N. benthamiana* leaves, two *Agrobacterium* cultures, each harboring one of the overexpression plasmids (individual OD600 at 0.8), were mixed in equal ratio and co-infiltrated to *N. benthamiana* leaves. Leaves were harvested five days after infiltration and were subjected to metabolite extraction and HPLC-QTOF-HRMS analysis as described above.

Constructs of CRISPR-SbSOMT and *Sorghum* Transformation

[0115] The CRISPR-SbSOMT construct was chemically synthesized (Gene Universal, USA). In brief, two gRNAs, including gRNA1: GTTGAACACGGTGTTCACG and gRNA2: GCACCGGACTACGCTGTGCG, were designed using CHOPCHOP v3^{sup.70}, individually introduced to the downstream of SbU6 promoters, SbU62.3 and SbU63.1, respectively, and were further cloned into the plasmid that harbors a selective marker (NPTII) to generate CRISPR-SbSOMT construct as described previously^{sup.50}. The CRISPR/Cas9 plasmid, pBUN411, was modified for *Sorghum* particle bombardment^{sup.71}. *Sorghum* tissue culture, transformation, and CRISPR-Cas9-mediated genome editing were performed as described previously^{sup.49}. Briefly, *Sorghum* inbred line Tx430 were grown in a temperature-controlled (18-28° C.) glasshouse to provide the initial explant for transformation. Immature seeds were harvested 11-15 days post anthesis. Immature embryos were isolated and placed on callus induction medium for generating embryogenic calli. The CRISPR-SbSOMT construct and pBUN411 plasmid were co-transformed into embryogenic calli by particle bombardment as described previously^{sup.72}. After transformation, potential transgenic plantlets were grown in a temperature-controlled (18-28° C.) glasshouse. Genome-edited plants (T0 and T1) were identified by PCR and direct sequencing using primers listed in Table 11.

X-Ray Crystallography

[0116] SbSOMT was prepared at 3.5 mg ml⁻¹ and pre-equilibrated with 0.5 mM of stilbene ligand in a final buffer of 20 mM HEPES pH 7.9, 150 mM NaCl, 4.5% dimethyl sulfoxide (DMSO) and 1 mM TCEP. Co-crystallization of SbSOMT and resveratrol were initially screened against commercial crystallization screens (Hampton Research, USA and QIAGEN, USA) with a sitting drop vapor diffusion approach incubated at 18° C. Two crystallizing conditions were further optimized. Crystals of SbSOMT-resveratrol- β -NAD ternary complex were obtained after two days of growing from the setup of 1 μ l of 5 mg ml⁻¹ SbSOMT, 0.15 μ l of 50 mM β -NAD and 0.5 μ l of reservoir solution (0.1 M sodium acetate pH 4.6, 0.2 M sodium acetate, 0.2 M NH₄Cl, 2.5% (w/v) polyethylene glycol 4000). SbSOMT co-crystallized with pinostilbene, piceatannol, or pterostilbene were obtained under the same condition. Crystal of SbSOMT-resveratrol binary complex was

initially obtained after 2-3 weeks of growing from the setup of 1 μ l of 5 mg ml⁻¹ SbSOMT and 0.5 μ l of reservoir solution (0.1 M MES pH 6.5, 0.2 M ammonium sulphate, 30% (w/v) polyethylene glycol monomethyl ether 5000). Subsequent crystallization was accelerated by seeding and crystals were harvested after 1 week. Crystals collected for X-ray diffraction was cryoprotected by 20% (v/v) glycerol topped onto the reservoir solution.

[0117] X-ray diffraction data were collected from BL19U1 beamline (wavelength at 0.979 Å) at the Shanghai Synchrotron Radiation Facility and processed by either XDS or HKL3000 package.sup.73-74. Data were analysed using pipelines provided in CCP4 suite.sup.75. Data reduction was performed using AIMLESS.sup.76. The initial structure of SbSOMT-resveratrol- β -NAD ternary complex was solved by molecular replacement; first through PHASER using a chimeric search model generated by MrBUMP, followed by model rebuilding from phase solution through BUCCANEER.sup.77-81. Following structures were solved by molecular replacement with PHASER using protomer of SbSOMT-resveratrol- β -NAD ternary complex as a search model. The built models were iteratively refined through automated REFMAC5 refinement and manual refinement in COOT.sup.81, 82. Validation and assessment of structure quality were performed using MolProbity and OneDep prior to finalization.sup.83, 84. The statistics and metrics of reported structure are compiled in Table 13. Figures featuring protein structure were prepared using UCSF Chimera.sup.85 or UCSF ChimeraX.sup.86.

Ligand Docking with AutoDock Vina

[0118] Ligand docking was performed using the default setting.sup.87. The determination of grid center was guided by superimposition of SbSOMT-resveratrol- β -NAD ternary complex, which the grid center was set in the binding pocket where the superimposed resveratrol is observed. For piceatannol docking in SbSOMT, the grid size was set to 18 Å×16 Å×12 Å, covering the entity of superimposed resveratrol. For docking in SbCOMT, a grid size of 24 Å×24 Å×24 Å was used with the similar grid center to cover the entire binding pocket. All docking were performed using global searching exhaustiveness of 8 and top 5 docked conformations were recorded for further analysis.

Phylogenetic Analysis

[0119] Multiple sequence alignment was done by ClustalW and Clustal Omega with default configurations.sup.88-89. The unrooted phylogenetic trees were constructed by maximum likelihood method with 1,000 bootstrap replicates using MEGA X90.

In Vitro Fungitoxicity Assays

[0120] A conidial suspension of *Collectotrichum sublineola* isolate TX430BB at a concentration of 5×10⁴ spore ml⁻¹ was prepared in double deionized water. Stilbenes including resveratrol, pinostilbene, and pterostilbene were dissolved in 80% (v/v) methanol and supplemented to the potato dextrose agar at concentrations ranging from 0 (mock) to 50 μ M. Ten microliters of conidial suspensions were transferred onto the water agar plates, and 9 biological replicates were conducted for each concentration of stilbenes. These plates were sealed, incubated overnight at room temperature in darkness, followed by incubation under constant light for 3 days. Spore germination was observed under a Leica DM500 light microscope (Leica, USA) after incubation in dark. Mycelial growth, represented by the fungal colony diameter, was assessed after the 3-day incubation period under light using the same microscope and the ImageJ software.sup.91.

Accession Numbers

[0121] Protein sequences analyzed in this study and used in phylogenetic analysis can be found under accession numbers listed in Table 14.sup.65.

Data Availability

[0122] Atomic coordinates and structure factors for the crystal structures reported in this work were deposited to the Protein Data Bank under accession numbers: 7VB8 (SbSOMT-resveratrol- β -NAD ternary complex), 7WAQ (SbSOMT-resveratrol binary complex), 7WAR (SbSOMT-pinostilbene- β -NAD ternary complex) and 7WAS (SbSOMT-pterostilbene- β -NAD ternary complex).

Results

Accumulation of O-Methylated Stilbenes in Pathogen-Infected *Sorghum*

[0123] The metabolic profiles of wild-type *Sorghum* upon infection of *Colletotrichum sublineola* (FIG. 1A) were first examined. Two *Sorghum* genotypes, BTx623 and SC748-5, susceptible and resistant to *C. sublineola* infection respectively.sup.37-38, were analyzed. Accordingly, four stilbenes including resveratrol, piceid (resveratrol 3-O-glucoside), and resveratrol derivatives O-methylated at their A-ring: pinostilbene (3-O-methylated resveratrol), and pterostilbene (3,5-bis-O-methylated resveratrol) were identified in the infected *Sorghum* mesocotyls (FIGS. 1B-1F). In addition, flavones (apigenin, luteolin, chrysoeriol, triclin; Table 3) and 3-deoxyanthocyanidins (orange-red pigments; apigeninidin, luteolinidin, diosmetinidin; Table 4), which are known *Sorghum* phytoalexins, were detected.sup.37-38. Notably, the resistant genotype SC748-5 accumulated approximately 9-fold more pterostilbene than the susceptible genotype BTx623 starting from 48 h post infection (FIG. 1B & Table 5). Taken together, stilbene accumulation in SC748-5 was more rapid, and at larger quantities than BTx623 during *C. sublineola* infection.

Sorghum Pathogen-Inducible SbSOMT Catalyzes Pterostilbene Biosynthesis

[0124] To identify potential OMTs involved in pterostilbene biosynthesis in *Sorghum*, an in silico expression dataset for *Bipolaris sorghicola*-infected *Sorghum* leaves (internet site matsui-lab.riken.jp/morokoshi/).sup.39-40 and in-house transcriptome dataset for *C. sublineola*-infected *Sorghum* mesocotyls.sup.41 were analyzed. Transcripts of SbOMT1 and SbOMT3, which were reported to methylate resveratrol in vitro and in transgenic plants, were not detected in either dataset (FIG. 8A). Meanwhile, SbCOMT, which encodes a bona fide canonical COMT.sup.18, 21, 22, is constitutively expressed in *Sorghum* (FIG. 8A). Furthermore, two putative OMT genes, SbSOMT (Sb07g004710) and SbOMT4 (Sb07g004690), showed pathogen-inducible expression patterns similar to that of SbSTS1 (Sb07g004700) (FIG. 8A) encoding stilbene synthase (STS) which generates resveratrol from p-coumaroyl-CoA and malonyl-CoAs (FIG. 1A).sup.5. Gene expression analyses confirmed their transcriptional upregulation in both *Sorghum* genotypes shortly after infection (FIGS. 8B-8D). In particular, the resistant genotype SC748-5 showed stronger expression for these genes than the susceptible genotype BTx623, consistent with its higher levels of pathogen-induced stilbene accumulation.

[0125] Subsequently, recombinant proteins expressed in *E. coli* were purified for in vitro enzyme assays. Both SbSOMT and SbCOMT generated pinostilbene and/or pterostilbene when incubated with resveratrol albeit at significantly different levels. Notably, SbSOMT converted about 60% of resveratrol into pterostilbene whereas SbCOMT displayed poor resveratrol 3,5-bis-O-methylation activities (FIG. 2A). Meanwhile, SbOMT4 showed minimal SOMT activities (FIGS. 9A-9C) and is thus unlikely to contribute to pterostilbene production in *Sorghum*. The catalytic activities of SbSOMT and SbCOMT towards hydroxycinnamic acids were also compared. Intriguingly, SbSOMT failed to methylate caffeic acid and 5-hydroxyferulic acid, whereas SbCOMT efficiently converted them to ferulic acid and sinapic acid, respectively (FIGS. 2B-2C).

[0126] Kinetic parameters of SbSOMT and SbCOMT were then determined using pinostilbene as a substrate since SbSOMT rapidly converts resveratrol into pterostilbene, thus preventing accurate quantification of pinostilbene produced. Results revealed the superior catalytic performance (kcat/Km and specific activity) of SbSOMT over SbCOMT and SbOMT4 towards pinostilbene, although SbSOMT showed a higher Km value (Table 1). These data indicated that SbSOMT, but not SbCOMT, is the primary SOMT for pterostilbene biosynthesis in *Sorghum*.

[0127] The biochemical function of SbSOMT was then examined via transient co-overexpression in *Nicotiana benthamiana* (FIGS. 10A-10D). Overexpression of both SbSTS1 and SbSOMT produced pterostilbene as the only stilbene product in agro-infiltrated leaves. By contrast, overexpression of SbSTS1 alone generated resveratrol and a small amount of pinostilbene, presumably due to endogenous promiscuous activities of tobacco OMT(s). These data are supportive for the role of SbSOMT in 3,5-bis-O-methylation of resveratrol to pterostilbene in planta.

Sorghum SbSOMT Mutants are Depleted in Pathogen-Inducible Pinostilbene and Pterostilbene [0128] *Sorghum* sbsomt mutants were generated via CRISPR/Cas9-mediated genome editing. Three homozygous sbsomt mutant lines (sbsomt-a, sbsomt-b1, sbsomt-b2; T1 generation) harboring different mutation pattern were isolated for metabolite analysis (FIG. 3A & FIGS. 11A-11F). In all mutant lines, the indels on exon 1 alone were sufficient to induce frameshift mutations and premature translation termination, resulting in knockout mutations (FIG. 12).

[0129] Metabolite profiles of sbsomt mutants were analyzed upon *C. sublineola* infection. (FIGS. 3B-3E & Table 6). Consistent with the results above (FIGS. 1C-1F), wild-type Tx430 mesocotyls accumulated resveratrol, piceid, pinostilbene and pterostilbene 72 h after infection. By contrast, the sbsomt-a/b1/b2 mutant mesocotyls accumulated resveratrol and piceid but not pinostilbene or pterostilbene even at 96 h after infection. These results established the indispensable role of SbSOMT in 3,5-bis-O-methylation of resveratrol to form pinostilbene and pterostilbene successively in *Sorghum*.

Sorghum Convergently Recruited SbSOMT Via Neofunctionalization of COMT

[0130] It has been discovered that SOMTs in grapevine (VvROMT) and Scots pine (PsPMT2) evolved convergently from non-COMT ancestors.^{sup.27, 28} Multiple sequence alignment showed that SbSOMT shares only 31.9% and 29.9% sequence identity with VvROMT and PsPMT2, respectively (FIG. 13). Instead, SbSOMT is phylogenetically related to canonical COMTs in Poaceae (FIG. 2D). In addition, an SbSOMT ortholog (ShSOMT) is found in Johnsongrass (*Sorghum halapense*; FIG. 2D). Meanwhile, no potential SbSOMT orthologs could be retrieved from wild sugarcane (*Saccharum spontaneum*) proteome (internet site plants.ensembl.org/; all homologous proteins with <50% identity; FIG. 14) although *Sorghum* spp. and *Saccharum* spp. share a recent common ancestor (both belonging to the same subtribe). Furthermore, ShOMT1, ShSOMT, SbOMT1, SbOMT4, SbSOMT form a *Sorghum*-specific clade sister to Poaceae naringenin 7-O-methyltransferases (NOMTs) (FIG. 2D). Hence, these OMTs were most likely recruited via duplication and neofunctionalization of an ancestral grass COMT. Overall, SbSOMT represents an independent and genus-specific acquisition of SOMT activities distinct from those in grapevine and Scots pine.

Substrate Binding Affinity and Regioselectivity of SbSOMT and SbCOMT

[0131] So far, the results presented herein revealed that SbSOMT catalyzes the 3,5-bis-O-methylation of resveratrol upon fungal infection. Meanwhile, SbCOMT efficiently methylates hydroxycinnamic acids but demonstrates minimal SOMT activities towards resveratrol and pinostilbene which might be attributed to poor substrate binding. However, isothermal titration calorimetry (ITC) study did not corroborate this possibility (Table 2 & FIG. 15). In fact, SbCOMT exhibited strong, micromolar binding affinities (K_d) towards resveratrol (1.55 μM), pinostilbene (1.79 μM), and pterostilbene (3.99 μM), comparable to those towards hydroxycinnamic acids.^{sup.21} Meanwhile, SbSOMT also showed strong binding affinities towards resveratrol and pinostilbene (4.69 μM and 2.86 μM respectively) but a higher K_d to pterostilbene (11.20 μM), demonstrating SbSOMT favors its substrates over product for binding. The SAM binding affinity of SbSOMT was unexpectedly weak ($K_d=90.00 \mu\text{M}$), whereas SbCOMT showed a K_d of 13.00 μM which is consistent with a previous study.^{sup.21} Collectively, SbSOMT and SbCOMT showed comparable stilbene-binding affinities, which would not account for their different catalytic activities towards resveratrol and pinostilbene.

[0132] Canonical COMTs including SbCOMT utilize the para-hydroxyl group of phenylpropanoids for proper substrate orientation.^{sup.21, 22} which concurs with results obtained herein of SbCOMT-catalyzed O-methylation of caffeic acid and 5-hydroxyferulic acid (FIG. 2B). On the other hand, SbSOMT-catalyzed O-methylation of resveratrol occurs in the A-ring which is not para-hydroxylated. Piceatannol (3'-hydroxylated resveratrol) was then used as an enzyme substrate since its stilbene B-ring is structurally identical to the phenolic ring in caffeic acid while its A-ring is identical to that in resveratrol (FIG. 4A). Both enzymes catalyzed in vitro O-methylation of

piceatannol, but occurring in a regioselective manner (FIG. 4B-4C). SbSOMT methylated the 3- and 5-hydroxyl groups on the piceatannol A-ring to produce 3'-hydroxypinostilbene and 3'-hydroxypterostilbene successively. By contrast, SbCOMT converted a substantial amount of piceatannol (ca. 60%) into isorhapontigenin by 3'-O-methylation in the B-ring with a conversion rate similar to those of SbCOMT-caffeic acid and SbSOMT-resveratrol reactions (FIG. 2A-2B). A small amount of isorhapontigenin was further methylated at its A-ring to 3'-methoxypinostilbene by SbCOMT (FIG. 4B-4C) and the conversion rate was similar to that of SbCOMT-resveratrol reaction (FIG. 2A). Furthermore, ITC experiments showed that SbCOMT displayed significantly stronger binding affinity towards piceatannol when compared to SbSOMT (Table 2; $K_d=1.26\ \mu\text{M}$ and $37.50\ \mu\text{M}$, respectively). Overall, these experiments strongly demonstrated the different regioselective O-methylation properties of SbSOMT and SbCOMT.

Global Structure of SbSOMT Resembles that of SbCOMT

[0133] To rationalize their different catalytic regioselectivities, structural analyses involving X-ray crystallography and computational methods were conducted. The crystal structure of an SbSOMT-resveratrol-nicotinamide adenine dinucleotide (β -NAD) ternary complex was solved (diffracted to $1.72\ \text{\AA}$ resolution, FIG. 4D; β -NAD was utilized as an additive to improve crystallization.^{sup.42}), an SbSOMT-resveratrol binary complex, and two ternary complexes reproduced with pinostilbene or pterostilbene (diffracted in the range of 2.10 - $2.56\ \text{\AA}$, FIG. 16A). All complexes were in high structural resemblance and depicted as a homodimer with an open conformation. Superimposition of SbSOMT and SbCOMT (PDB: 4PGH.^{sup.21}) revealed their high structural similarity (RMSD= $2.322\ \text{\AA}$), except that SbSOMT harbors an extended loop (Val103-Cys114) that forms a threonine-rich dimerization interface (FIG. 4D).

Structural Analyses and in Silico Docking Support Different Stilbene Orientations within the Substrate Binding Pockets of SbSOMT and SbCOMT

[0134] Close examination of the SbSOMT substrate binding pocket within the SbSOMT-resveratrol- β -NAD complex provided further insights into the interactions between SbSOMT and its stilbene substrates. The pocket mainly composed hydrophobic residues, and the stilbene backbone was secured by hydrophobic interactions with Met143, Ile144, Met175, Phe189, Met193, Trp279, Tyr311, Leu329, Met333, Thr336, in addition to Leu28 of the adjacent protomer (FIG. 5A). The apparent binding cavity was confined to fit the stilbene backbone in a planar configuration with minimal rotational flexibility. Of the three hydroxyl groups on resveratrol capable of forming hydrogen bonds, only the methyl-accepting 3-OH group (A-ring) formed direct hydrogen bonds to SbSOMT residues (FIG. 5B), including the catalytic residues His282 ($2.80\ \text{\AA}$) and Asp283 ($3.26\ \text{\AA}$). Other hydroxyl groups were shielded by water molecules and did not directly interact with SbSOMT. The same binding mode was adopted for pinostilbene or pterostilbene in the corresponding SbSOMT co-crystals (FIG. 5C-5D & FIG. 16B). Overall, these structures depicted a productive substrate orientation within the substrate binding pocket which greatly favors A-ring O-methylation. Following 3-O-methylation of resveratrol, the pinostilbene intermediate will need to be dissociated and re-inserted in order to re-position the A-ring for 5-O-methylation to produce pterostilbene. Similarly, in silico docking of piceatannol with SbSOMT (FIG. 5E) conformed to such substrate orientation, with the A-ring positioned in proximity to the catalytic residues of SbSOMT, hence favoring the 3,5-bis-O-methylation to produce 3'-hydroxypinostilbene and 3'-hydroxypterostilbene (FIG. 4B-4C).

[0135] Docking stilbenes with SbCOMT (PDB: 4PGH.^{sup.21}) revealed a contrasting substrate orientation to that of SbSOMT, suggesting that SbCOMT preferentially positions stilbenes with the B-ring close to the catalytic residues and the A-ring reaching inwardly into the pocket. SbCOMT-piceatannol docking showed a single favorable orientation of piceatannol among five hits which were docked at a sub-optimal position (FIG. 5E). In the top hit, the B-ring 3'- and 4'-hydroxyl groups were stabilized by hydrogen bonds with Asn323 while A-ring 5-hydroxyl group interacted with Asn128 (FIG. 5E). Consistent with the enzyme assay results (FIG. 2A), the 3'-hydroxyl group

on the B-ring was orientated close to the catalytic residues and would thus predominantly favors the methylation of 3'-hydroxyl group (B-ring) over 3-/5-hydroxyl groups (A-ring) (FIG. 5E). Additionally, docking resveratrol with SbCOMT revealed two opposite orientations with similar free energies of binding (FIG. 5F), representing two competing binding modes which hampered its catalytic performance (FIG. 2A). In fact, the non-productive orientation with the B-ring closer to the catalytic residues was more energetically favorable (-7.3 kcal/mol, in blue), while the productive orientation with A-ring closer to the catalytic residues was slightly less stable (-7.1 kcal/mol, in grey). Collectively, these analyses fully corroborated with results of the biochemical assays and stilbene profiles of *Sorghum* presented herein, and provided a mechanistic rationale for the apparent differences in catalytic activity between SbCOMT and SbSOMT. Additionally, as the methyl-accepting —OH groups in piceatannol and hydroxycinnamic acids are both adjacent to a para-OH group (i.e., 4'-OH in piceatannol and 4-OH in hydroxycinnamic acids), this may be a prerequisite for efficient O-methylation by COMTs as suggested previously.^{sup.21, 22.}

Sbsomt and SbCOMT Employ the Same Catalytic Residues but Different Substrate Binding Orientations

[0136] The catalytic residues of SbSOMT are highly conserved with all reported COMTs, including His282 as the key nucleophile, whereas Asp283, Glu310, and Glu342 collectively charge the basicity or confine His282 in its reactive conformation (FIG. 5B).^{sup.21, 22, 43, 44.} Accordingly, the surrounding electrostatic force allows NF of His282 to deprotonate the methyl-accepting —OH group, followed by a SN2 nucleophilic attack of the methyl-donor group on SAM by the oxyanion (—O—) (FIG. 17).^{sup.21, 22, 43, 44.} The roles of His282 and Asp283 have been validated by mutagenesis, but not for those of Glu310 and Glu342 (PDB: 4PGG.^{sup.21,} 5ICG.^{sup.45,} 1KYZ.^{sup.44,} and 6I5Z.^{sup.46}). Here, H282A, D283A, E310A, and E342A mutant proteins of SbSOMT were generated for catalytic and binding assays with resveratrol. H282A mutant protein was catalytically defective and unstable for ITC study (FIGS. 6A-6B). To further elucidate the role of His282 in resveratrol binding (FIG. 5B), a stable SbSOMT H282N/D283A double mutant protein was generated. While the D283A protein showed a slightly stronger affinity towards resveratrol ($K_d=1.31$ μ M), the H282N/D283A mutant protein showed a significantly weaker affinity ($K_d=32.90$ μ M), implying that His282 plays a moonlighting role in substrate binding in SbSOMT in addition to catalysis (Table 7). Meanwhile, D283A and E342A mutations resulted in significant reduction in pinostilbene yield by 94.7% and 91.9%, respectively (FIG. 6A). In addition, both of them failed to generate pterostilbene when incubated with resveratrol despite that they showed considerably stronger affinities to resveratrol (FIG. 6B & Table 7). Unexpectedly, E310A mutation only caused reduction of pterostilbene yield (by 70.1%), accompanying with reshuffled stilbene binding affinities (FIG. 6B & Table 7). It was reasoned that the E310A mutation decreased the rigidity and basicity of His282, thus weakening SbSOMT selectivity towards the reactive hydroxyl group against the inactive methoxy group, which is a vital feature allowing the subsequent O-methylation of the same stilbene substrate (FIGS. 5B-5D).

[0137] In SbCOMT, two key amino acid residues, Asn128 and Asn323, cooperatively mediate the productive orientation for catalysis.^{sup.21.} Docking piceatannol with SbCOMT revealed that Asn323 forms a hydrogen bond with the para-OH group on the stilbene B-ring whose meta-OH group is positioned optimally for methylation, while Asn128 likely serves to stabilize their interaction via another hydrogen bond with 5-OH on the non-para-hydroxylated stilbene A-ring (FIG. 5E). In striking contrast, the equivalent positions in SbSOMT are occupied by two highly hydrophobic residues, Ile144 and Phe337, which are unfavorable for interactions with the para-OH group of an aromatic ring as predicted by in silico docking (FIG. 5E). On the other hand, these residues may contribute to accommodate the stilbene A-ring, which is not para-hydroxylated, for methylation of its meta-hydroxyl groups. I144N, F337N and I144N/F337N SbSOMT mutant proteins were then generated for catalytic and binding assays. All three SbSOMT mutant proteins showed significantly weakened binding for resveratrol, with F337N and I144N/F337N mutant

proteins displaying undetectable binding affinities (Table 7). Correspondingly, both I144N and F337N produced substantially less pterostilbene (24.8% and 97.7% respectively), and accumulated slightly more pinostilbene when compared to the WT protein (28.4% and 50.8% respectively; FIGS. 6C-6D). Meanwhile, the I144N/F337N mutant protein showed complete abolishment of SOMT activities (FIGS. 6C-6D). As pterostilbene represents the major product generated by SbSOMT-resveratrol reactions (FIG. 2A), these results established the important role of Ile144 and Phe337 in the resveratrol binding mechanism adopted by SbSOMT.

A Canonical COMT Likely Catalyzes Isorhapontigenin Biosynthesis in Wild Sugarcane
[0138] *Saccharum* spp. were reported to accumulate resveratrol and piceatannol (3'-hydroxylated resveratrol) upon mechanical wounding or fungal infection.^{sup.2, 36} (FIG. 7A). As *Saccharum* spp. are Saccharinae grasses like *Sorghum* spp., it was investigated whether O-methylated derivatives could also be detected in wild sugarcane whose genome is available.^{sup.47} The slender stalks of flowering wild sugarcane (*S. spontaneum*) were sawn into 2-cm segments (excluding nodes) for metabolic profiling. No stilbenes were detected in freshly-cut segments whereas resveratrol, piceatannol, and isorhapontigenin (3'-O-methylated piceatannol) started to accumulate 72 h after wounding (FIG. 7B, FIG. 18A-18C, and Table 8). A miniscule amount of a hexoside of piceatannol was also tentatively identified (FIGS. 19A-19B). Overall, mechanically-wounded wild sugarcane produced piceatannol and its B-ring O-methylated derivative isorhapontigenin which constituted a small portion of the stilbene profile (FIG. 7B), while no A-ring O-methylated stilbene could be detected. Next, the understanding of isorhapontigenin biosynthesis was pursued in wild sugarcane. Substantial upregulation of SsSTS gene expression (Sspon.06G0010290-2P) 72 h after wounding was detected (FIG. 20A, FIG. 20C & FIG. 21). Under in vitro condition, recombinant SsSTS generated piceatannol with caffeoyl-CoA as the starter substrate but failed to generate isorhapontigenin when feruloyl-CoA was used (FIGS. 22A-22E). Hence, SsSTS is a functional STS but an OMT is most likely required for 3'-O-methylation of piceatannol to produce isorhapontigenin. Meanwhile, a canonical SsCOMT (Sspon.06g0010980-3C) highly conserved with SbCOMT (>94.0% protein sequence identity) was identified in wild sugarcane (FIG. 14 & FIG. 23). It also harbors the key Asn residues corresponding to Asn128 and Asn323 in SbCOMT (FIG. 23). In fact, recombinant enzyme assays demonstrated that SsCOMT and SbCOMT showed the same catalytic regioselectivity. Accordingly, isorhapontigenin, ferulic acid, and sinapic acid were generated when SsCOMT was incubated with piceatannol, caffeic acid, and 5-hydroxyferulic acid respectively (FIG. 7C & FIGS. 24A-24F). However, only small amounts of pinostilbene were generated when resveratrol was used as substrate (FIGS. 24G-24I). In addition, SsCOMT shares similar substrate-binding properties with SbCOMT (Table 9). Gene expression further revealed that SsCOMT is constitutively expressed in the wild sugarcane segments despite its downregulation after wounding (FIG. 20B & FIG. 20D). Taken together, SsCOMT is an important candidate for isorhapontigenin production through 3'-O-methylation of piceatannol in the B-ring.

Discussion

[0139] O-Methylated stilbenes are well-acclaimed for their health-promoting benefits and exceptional bioavailability.^{sup.35-48} *Sorghum* is a major staple crop and wild sugarcane is a genetic resource for sugarcane breeding. In this study, the potential of these Saccharinae grasses as biofactories for O-methylated stilbenes as well as the OMTs for regioselective stilbene O-methylations were revealed. The findings presented herein also provide insights into bioengineering of specific O-methylated stilbenes via molecular breeding and transgenic approaches, representing a unique opportunity to improve dietary intake of these nutraceuticals which are scarcely present in natural food sources. Recent breakthroughs in *Sorghum* biotechnology, including stable transformation and

[0140] CRISPR/Cas9-mediated genome editing as well as its metabolic versatility, have warranted *Sorghum* as an emerging model for investigation and manipulation of specialized metabolism.^{sup.49-51} Here, complete depletion of O-methylated stilbenes in infected *Sorghum*

SbSOMT CRISPR-Cas9 mutants (FIGS. 3A-3E) firmly established SbSOMT as the primary SOMT for pathogen-inducible pterostilbene biosynthesis. Concomitantly, molecular, biochemical, and structural characterizations were integrated to elucidate the mechanistic details of SbSOMT-stilbene reactions. The first crystal structure of a bona fide stilbene O-methyltransferase, SbSOMT is reported herein. The enzyme utilizes the His282-Asp283-Glu310-Glu342 catalytic residues to generate 3,5-bis-O-methylated stilbenes (FIG. 2A, FIGS. 4B-4C, FIGS. 5B-5F & FIGS. 16A-16C). The same catalytic residues are found in canonical COMTs, indicating the highly conserved O-methylation catalysis irrespective of phenolic substrates involved^{13, 21, 22, 52, 53}. It was demonstrated that Glu310 positions the 5-OH group, instead of the more hydrophobic 3-OCH₃ group, in close proximity to His282 for the second O-methylation (FIGS. 5C-5D & FIG. 16B). As Glu310 is highly conserved in most OMTs (with some harboring an Asp), this feature may represent a common mechanism for mediating successive O-methylations of structurally different phenolic substrates.

[0141] Meanwhile, combinatorial analyses presented herein rationalized the regioselectivities of SbSOMT and canonical COMTs underlain by specific substrate binding modes instead of alterations in catalysis or substrate affinity (FIG. 2A, FIGS. 4B-4C, FIGS. 5B-5F, FIGS. 6A-6D, FIGS. 7A-7C, Table 1, Table 2, FIG. 17 & Table 7). The hydrophilic Asn128/Asn323 residues in SbCOMT, which are highly conserved in COMTs from diverse plant lineages, orchestrate productive coordination with their native substrates, i.e. hydroxycinnamic acids and their analogs, via hydrogen bonding.^{sup.21, 22} The same binding mechanism is apparently crucial for attaining productive and energetically favorable orientation of piceatannol inside the binding pocket of Saccharinae COMTs including SbCOMTAsn128/Asn323 and SsCOMTAsn130/Asn323 for B-ring O-methylation (FIG. 5E & FIGS. 24A-24B). In marked comparison, the recruitment of more hydrophobic residues in SbSOMT like Ile144 and Phe337 during its emergence in the *Sorghum* genus would considerably favor hydrophobic interactions with the C4 and C3' of resveratrol (and piceatannol) and pinostilbene (and 3'-hydroxypinostilbene), hence favoring A-ring O-methylation (FIGS. 4B-4C, FIG. 5E & FIGS. 6C-6D).

[0142] Thorough mining of the Protein Data Bank (PDB) retrieved a diverse panel of 24 OMT-ligand complexes (from 13 OMTs with $\geq 30\%$ protein identity to SbCOMT/SbSOMT), supporting a general occurrence of compatible polarity pairing between the amino acid residue equivalent to SbCOMTAsn323/SbSOMTPhe337 in OMTs and their substrates (Table 10). It was reasoned that upon conformational transitions, this residue and the catalytic residues are brought into close proximity, leading to compatible polarity pairing between 12 (out of 13) OMTs and their respective substrate (FIG. 25 & FIGS. 26A-26X). For example, six OMTs harboring an Asn residue at this position is paired to a polar-OH or -OCH₃ moiety of the ligand (Table 10 & FIGS. 26A-26X).^{sup.43, 44, 46, 54-57} For the others, the equivalent residue is a more hydrophobic residue (Thr, Val or Phe) which is paired to a non-polar group such as a methyl group or an aromatic ring of the ligand (Table 10 & FIGS. 26A-26X). Consistently, two previously characterized SOMTs, VvROMTPhe318 and SbOMT3-Ile336, harbor a hydrophobic residue at this equivalent position.^{sup.28, 29} Meanwhile, naringenin OMTs, which harbor a hydrophobic Leu residue (OsNOMTLeu335; SbNOMTLeu324; ZmNOMTLeu325) paired to the aromatic carbon (C6), catalyze 7-O-methylation (A-ring) of naringenin to generate sakuranetin, but do not utilize hydroxycinnamic acid substrates.^{sup.58} Collectively, the polarity of the amino acid residue in OMTs equivalent to SbCOMTAsn323/SbSOMTPhe337 may govern substrate selectivity through a compatible polarity pairing with the functional group vicinal to the methyl-accepting —OH group in a substrate (FIG. 25 & FIGS. 26A-26X). Consequently, replacement of Asn323 by a hydrophobic residue represents a key molecular event towards the functional divergence of regioselective OMT activities from canonical COMTs in plants. On the other hand, polarity pairing between ligand and SbCOMTAsn128 is only observed in some OMT-ligand complexes, as the equivalent residues in OMTs are diversified (Table 10). Overall, the findings presented herein

underpin subsequent bio-engineering of OMTs, either by directed evolution or targeted site-directed mutagenesis, for production of regioselective O-methylated phenolic nutraceuticals/pharmaceuticals in vitro or in planta.

[0143] *Sorghum* and wild sugarcane are closely-related species but resveratrol is their only common stilbene detected by experimental conditions used herein. In addition to resveratrol, *Sorghum* produces pinostilbene and pterostilbene upon fungal infection while wild sugarcane accumulates resveratrol, piceatannol, and isorhapontigenin following mechanical wounding (FIGS. 1A-1F & FIGS. 7A-7C). The lack of B-ring O-methylated stilbenes (pinostilbene and pterostilbene) in wounded wild sugarcane stems primarily from the absence of SbSOMT orthologs, which are unique to *Sorghum* spp. On the other hand, the presence of isorhapontigenin (B-ring O-methylated) is resulting from canonical SsCOMT activities although genetic evidence is required for confirmation. Meanwhile, the underlying causes for the exclusivity of piceatannol (3'-hydroxylated) and related stilbenes in *Saccharum* spp. within the Saccharinae subtribe remain elusive. Elucidation of the piceatannol biosynthetic pathway, in combination with the aforementioned bio-engineering of OMTs for targeted stilbene O-methylation, would further unleash the potential of these Saccharinae crops for stilbene biofortification.

[0144] Apparently, *Sorghum*-specific SOMTs was originated from the canonical and ubiquitous COMTs in Poaceae (FIG. 2D). The COMT-to-SOMT evolution fits well into the generally-accepted Yeas-Jensen model which suggests that a promiscuous enzyme with poor activities towards specific substrates often serves as the basic scaffold undergoing sequence divergence (e.g. the aforementioned replacement of SbCOMTAsn323 by SbSOMTPhe337) to optimize catalytic efficiency^{59, 60}. Further nonsynonymous mutations within an ancestral *Sorghum* COMT allowed neofunctionalization with acquisition of efficient resveratrol 3,5-bis-O-methylation activities in SbSOMT (FIG. 2A). Such molecular evolution might have been driven by the superior potency of pterostilbene over resveratrol and pinostilbene as a broad-spectrum phytoalexin (FIG. 27A-27M).^{sup.6, 61} Both SbSOMT and OsNOMT have lost the ability to O-methylate hydroxycinnamic acids (FIG. 2B), which are key in planta substrates of COMTs.^{sup.21, 22} Further investigations will shed new light on whether compromised substrate recognition pattern is necessary, and to what extent, to achieve the highly efficient SOMT activities in SbSOMT. Since both SbSOMT and SbCOMT utilize the same catalytic residues and mechanism for O-methylation, evolution of pterostilbene biosynthesis primarily hinged on optimizing the stilbene binding mode with productive orientations. Furthermore, the recruitment of SbSTS1 from a chalcone synthase scaffold for resveratrol biosynthesis would logically predate that of SbSOMT from SbCOMT⁵. An increasing number of specialized metabolic pathways are considered to have evolved in such order as gene clusters.^{sup.62-64} Correspondingly, physically-linked SbSTS1 (Sb07g004700), SbOMT4 (Sb07g004690), and SbSOMT (Sb07g004710) (FIG. 28) were identified as constituents of a gene cluster for specialized metabolism (Cluster 408) in the *Sorghum* genome.^{sup.63-65}

TABLE-US-00001 TABLE 1 Kinetic parameters of SbSOMT and SbCOMT with pinostilbene as a substrate. SbCOMT SbSOMT K_{sub.m} (μM) 3.14 ± 1.40 4.62 ± 0.46 k_{sub.cat} (min.^{sup.-1}) 0.14 ± 0.01 10.30 ± 0.20 k_{sub.cat}/K_{sub.m} (min.^{sup.-1} μM.^{sup.-1}) 0.045 2.229 Specific activity (nmol 3.46 ± 0.27 290.04 ± 5.64 min.^{sup.-1} mg.^{sup.-1}) Values refer to means ± SD (n = 3).

TABLE-US-00002 TABLE 2 Dissociation constants of SbSOMT, SbCOMT with stilbenes. K_{sub.d} (μM) SbCOMT SbSOMT Resveratrol 1.55 ± 0.13 4.69 ± 0.54 Pinostilbene 1.79 ± 0.09 2.86 ± 0.33 Pterostilbene 3.99 ± 0.24 11.20 ± 2.23 Piceatannol 1.26 ± 0.06 37.50 ± 4.33 SAM 13.00 ± 0.83 90.00 ± 9.59 Results are expressed as means of K_{sub.d} ± standard error derived from curve fitting.

TABLE-US-00003 TABLE 3 High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of flavone aglycones in control or *Colletotrichum sublineola*-infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes. BTx623 SC748-5 Time (h) Control Infected Control Infected Apigenin [peak area (cps × 10.^{sup.6}) g.^{sup.-1}

FW] 24 4.08 ± 0.19 5.48 ± 4.58 3.95 ± 0.57 4.10 ± 0.30 48 7.18 ± 2.34 12.41 ± 1.67 8.06 ± 0.83
 52.27 ± 12.73 72 12.25 ± 1.93 105.55 ± 31.98 12.24 ± 2.55 33.61 ± 9.98 96 13.11 ± 1.04
 272.56 ± 100.65 7.42 ± 1.71 20.59 ± 3.92 Luteolin [peak area (cps × 10.sup.6) g.sup.-1 FW] 24
 1.20 ± 0.25 0.11 ± 0.05 2.38 ± 0.57 2.57 ± 0.45 48 1.16 ± 0.11 1.66 ± 0.43 3.37 ± 1.35 158.69 ±
 29.51 72 1.62 ± 0.09 21.66 ± 10.97 4.94 ± 1.60 512.03 ± 65.46 96 5.35 ± 0.82 65.99 ± 32.20 7.03 ±
 2.30 497.52 ± 135.58 Chrysoeriol [peak area (cps × 10.sup.6) g.sup.-1 FW] 24 3.19 ± 2.60 4.66 ±
 0.87 4.75 ± 1.15 8.88 ± 0.86 48 6.18 ± 0.27 8.02 ± 0.64 9.54 ± 1.11 108.23 ± 14.69 72 10.26 ±
 1.15 28.49 ± 5.22 13.31 ± 1.97 308.20 ± 24.77 96 12.91 ± 1.07 75.03 ± 21.31 14.01 ± 4.62
 364.55 ± 100.89 Tricin [peak area (cps × 10.sup.7) g.sup.-1 FW] 24 2.58 ± 0.27 1.59 ± 1.17 41.14
 ± 9.33 35.36 ± 2.76 48 2.75 ± 0.55 2.62 ± 0.20 59.66 ± 10.28 75.39 ± 16.11 72 3.80 ± 1.03 4.52
 ± 0.53 88.92 ± 6.14 242.87 ± 31.12 96 4.58 ± 0.04 9.76 ± 1.42 73.18 ± 24.74 396.46 ± 22.25
 Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of
 individual peaks with those from authentic standards. Apigenin-d.sub.5 was used as an internal
 standard for quantitation. Values refer to means ± SD (n = 3). cps: count per second, FW: fresh
 weight.

TABLE-US-00004 TABLE 4 High-performance liquid chromatography-quadrupole time-of-flight
 MS (HPLC-QTOF-MS) analysis of 3-deoxyanthocyanidin aglycones in mock or *Colletotrichum*
sublineola-infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum
 genotypes. BTx623 SC748-5 Time (h) Control Infected Control Infected Apigeninidin [peak area
 (cps × 10.sup.8) g.sup.-1 FW] 48 n.d. 1.11 ± 0.25 n.d. 4.69 ± 1.65 72 n.d. 10.52 ± 5.66 n.d.
 10.93 ± 1.72 96 n.d. 19.37 ± 2.09 n.d. 9.49 ± 1.24 Luteolinidin [peak area (cps × 10.sup.6) g.sup.-
 1 FW] 48 n.d. 2.44 ± 0.65 n.d. 822.19 ± 271.66 72 n.d. 153.69 ± 94.92 n.d. 2735.93 ± 339.38
 96 n.d. 246.27 ± 45.66 n.d. 2403.59 ± 761.50 Diosmetinidin [peak area (cps × 10.sup.6) g.sup.-1
 FW] 48 n.d. 0.97 ± 0.12 n.d. 167.38 ± 78.93 72 n.d. 21.13 ± 12.19 n.d. 803.91 ± 213.56 96 n.d.
 46.81 ± 10.43 n.d. 1352.29 ± 185.15 Compound annotation was achieved by comparing the
 accurate mass and fragmentation pattern of individual peaks with those from authentic standards.
 Apigenin-d.sub.5 was used as an internal standard for quantitation. Values refer to means ± SD (n =
 3). cps: counts per second, FW: fresh weight, n.d.: not detected.

TABLE-US-00005 TABLE 5 High-performance liquid chromatography-quadrupole time-of-flight
 MS (HPLC-QTOF-MS) analysis of stilbenes in mock or *Colletotrichum sublineola*-infected
 mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes. BTx623 SC748-5
 Time (h) Control Infected Control Infected Resveratrol (µg g.sup.-1 FW) 72 n.d. 0.03 ± 0.01 n.d.
 0.31 ± 0.00 96 n.d. 0.47 ± 0.23 n.d. 0.91 ± 0.19 Piceid (µg g.sup.-1 FW) 72 n.d. 2.63 ± 0.23 n.d.
 7.99 ± 0.46 96 n.d. 6.27 ± 1.27 n.d. 19.60 ± 3.24 Pinostilbene (µg g.sup.-1 FW) 72 n.d. 0.05 ±
 0.00 n.d. 0.22 ± 0.00 96 n.d. 0.07 ± 0.02 n.d. 0.57 ± 0.14 Pterostilbene (µg g.sup.-1 FW) 48 n.d.
 n.d. n.d. 0.03 ± 0.02 72 n.d. 0.16 ± 0.06 n.d. 1.56 ± 0.13 96 n.d. 0.61 ± 0.23 n.d. 5.66 ± 1.89

Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of
 individual peaks with those from authentic standards. Apigenin-d.sub.5 was used as an internal
 standard for quantitation. Values refer to means ± SD (n = 3). FW: fresh weight, n.d.: not detected.

TABLE-US-00006 TABLE 6 High-performance liquid chromatography-quadrupole time- of-flight
 MS (HPLC-QTOF-MS) analysis of stilbenes in mock or *Colletotrichum sublineola*-infected
 mesocotyls of wild-type Tx430 and sbsomt mutants. Time (h) Tx430 sbsomt-a sbsomt-b1 sbsomt-
 b2 Resveratrol (µg g.sup.-1 FW) 72 0.09 ± 0.12 0.09 ± 0.02 0.10 ± 0.03 0.11 ± 0.04 96 0.18 ± 0.03
 0.31 ± 0.12 0.49 ± 0.07 0.53 ± 0.03 Piceid (µg g.sup.-1 FW) 72 1.60 ± 0.25 2.60 ± 1.20 1.78 ±
 1.06 1.97 ± 1.34 96 5.74 ± 0.70 8.03 ± 2.33 18.14 ± 2.32 17.14 ± 5.45 Pinostilbene (µg g.sup.-1
 FW) 72 0.03 ± 0.00 n.d. n.d. n.d. 96 0.09 ± 0.02 n.d. n.d. n.d. Pterostilbene (µg g.sup.-1 FW) 72
 0.13 ± 0.02 n.d. n.d. n.d. 96 1.84 ± 0.32 n.d. n.d. n.d. Compound annotation was achieved by
 comparing the accurate mass and fragmentation pattern of individual peaks with those from
 authentic standards. Apigenin-d.sub.5 was used as an internal standard for quantitation. Values refer
 to means ± SD (n = 3). FW: fresh weight, n.d.: not detected.

TABLE-US-00007 TABLE 7 Dissociation constant (K_{sub.d}) of SbSOMT mutants with stilbenes
 Stilbene SbSOMT mutant Stoichiometry, N K_{sub.d} (μM) Resveratrol I144N 0.6 151.6 ± 31.1
 D283A 0.58 ± 0.01 1.31 ± 0.13 E310A 0.36 ± 0.04 18.5 ± 2.58 F337N — No binding E342A 0.75
 ± 0.01 11.1 ± 0.75 I144N/F337N — No binding H282N/D283A 0.61 ± 0.17 32.90 ± 5.93
 Pinostilbene E310A 0.59 ± 0.01 3.91 ± 0.33 Pterostilbene E310A 0.40 ± 0.08 6.04 ± 2.08 Results
 are expressed as means of K_{sub.d} ± SE derived from curve fitting. N of I144N was fixed at 0.6
 (average to reported Ns in this table) to estimate the K_{sub.d}.

TABLE-US-00008 TABLE 8 High-performance liquid chromatography-quadrupole time-of-flight
 MS (HPLC-QTOF-MS) analysis of stilbenes in mechanically-wounded stalks wild sugarcane.
 Wounded wild sugarcane Resveratrol Piceatannol Isorhapontigenin Time (h) (μg g_{sup}.⁻¹ FW) (μg
 g_{sup}.⁻¹ FW) (μg g_{sup}.⁻¹ FW) 0 n.d. n.d. n.d. 24 n.d. n.d. n.d. 72 2.55 ± 1.55 9.37 ± 5.98 1.03 ±
 0.87 120 7.87 ± 3.03 33.79 ± 4.50 2.18 ± 1.77 Compound annotation was achieved by comparing
 the accurate mass and fragmentation pattern of individual peaks with those from authentic
 standards. Apigenin-d_{sub}.5 was used as an internal standard for quantitation. Values refer to means
 ± SD (n = 3). FW: fresh weight, n.d.: not detected.

TABLE-US-00009 TABLE 9 Dissociation constants of SsCOMT with stilbenes. K_{sub.d} (μM)
 SsCOMT Resveratrol 2.78 ± 0.33 Pinostilbene 5.57 ± 0.77 Pterostilbene 9.93 ± 4.04 Piceatannol
 25.50 ± 14.20 SAM 25.00 ± 12.90 Results are expressed as means of K_{sub.d} ± standard error
 derived from curve fitting.

TABLE-US-00010 TABLE 10 List of OMT-ligand complexes deposited to PDB and the equivalent
 residues of SbCOMT_{sup}.Asn128/SbSOMT_{sup}.Ile144 and
 SbCOMT_{sup}.Asn323/SbSOMT_{sup}.Phe337. Residue at equivalent positions SbCOMT SbCOMT
 Vicinal Asn128 Asn323 functional SEQ PDB SbSOMT SbSOMT Ligand group_{sup}.# and ID ID
 UNIPROT Ile144 Phe337 identity polarity NO Ref. 1FP1 P93324 Leu139 Val333 Substrate —H
 and non-polar 82 [43] 1KYW P28002 Asn131 Asn324 Substrate —OH and polar 83 [44] 1KYZ
 P28002 Asn131 Asn324 Product —OH and polar 1ZG3 Q29U70 Ala126 Phe326 Substrate —H
 and non-polar 84 [54] 1ZGA Q29U70 Ala126 Phe326 Substrate —H and non-polar 1ZGJ Q29U70
 Ala126 Phe326 Product —H and non-polar 2QYO Q06YR3 Val123 Val319 Canonical —H and
 non-polar 85 NA product 3I58 Q84HC8 Asp107 Phe294 Product —H and non-polar 86 [55] 3I5U
 Q84HC8 Asp107 Phe294 Substrate Not compared analogue (non- reactive) 3I64 Q84HC8 Asp107
 Phe294 Substrate Fused benzene ring analogue and non-polar (reactive) 3P9I Q9ZTU2 Asn128
 Asn321 Product —OH and polar 87 [22] 3P9K Q9ZTU2 Asn128 Asn321 Product —OH and polar
 3REO O04385 Ala134 Asn327 Substrate —OMe and polar 88 3TKY O04385 Ala134 Asn327
 Substrate —OMe and polar 5CVU O04385 Ala134 Asn327 Substrate —OMe and polar 5I2H
 D5STZ7 Ala110 Thr322 Non-productive Not compared 89 conformation 5ICE Q5C9L7 Ile114
 Thr311 Substrate —H and non-polar 90 5ICF Q5C9L7 Ile114 Thr311 Non-reactive Not compared
 inhibitor 5XOH A0A166U5H3 His126 Val320 Substrate Fused furan ring and 91 non-polar 6I6L
 I3V6A7 Thr157 Asn351 Product —OMe and polar 92 6I6M I3V6A7 Thr157 Asn351 Substrate —
 OMe and polar 6I72 Q9M602 Asn131 Asn324 Canonical —OH and polar 93 substrate _{sup}.#The
 vicinal functional group refers to the functional group vicinal to the reactive site positioned
 between catalytic dyads and SbCOMT_{sup}.Asn323/SbSOMT_{sup}.Phe337 equivalent position as depicted in
 FIG. 25 & FIGS. 26A-26X. NA, Not available (work to be published).

TABLE-US-00011 TABLE 11 Primers. Sequence (5' to 3') (restriction sites Primer
 name added are in *italic*) Purposes SbEIF4a-qRT-F CAACCTTTGTCACCCGCGATGA
 (SEQ qRT-PCR experiments for SbEIF4α ID NO: 36) SbEIF4a-qRT-R
 TCCAGAAACCTTAGCAGCCCA (SEQ ID NO: 37) SbSTS-qRT-F
 TGCTACGGTGTGGCCATT (SEQ ID qRT-PCR experiments for SbSTS1 NO: 38)
 SbSTS-qRT-R ATCGACTTGTGGCATATCCTCTT (SEQ ID NO: 39) SbOMT4-qRT-F
 CCTAAAGTGGATTCTTCACGATTG qRT-PCR experiments for SbOMT4 (SEQ ID
 NO: 40) SbOMT4-qRT-R GCTAGTTGCGATGTTAGTGTTTC (SEQ ID NO: 41)

SbSOMT-qRT-F AGGTGGTGGTTCGATAA (SEQ qRT-PCR experiments for
 SbSOMT ID NO: 42) SbSOMT-qRT-R ACGATCACCTTGCCTCTCAC (SEQ ID NO:
 43) SsGADPH-qRT-F TTGGTTTCCACTGACTTCGTT (SEQ ID qRT-PCR experiments
 for SsGADPH NO: 44) SsGADPH-qRT-R CTGTAGCCCCACTCGTTGT (SEQ ID NO:
 45) SsSTS-qRT-F AGAGCGAACACCTTACCGAC (SEQ ID qRT-PCR experiments for
 SsSTS NO: 46) SsSTS-qRT-R GCCGAGTACGAGCTCATGTT (SEQ ID NO: 47)
 SsCOMT-qRT-F GAGGACAAGGACGGCAAGTA (SEQ qRT-PCR experiments for
 SsCOMT ID NO: 48) SsCOMT-qRT-R ACCGCGTCCTTGAGGTAGTA (SEQ ID NO:
 49) SbOMT4-RE-F CGCGGATCCATGGGCAGCTATACTAC Cloning of SbOMT4 in
 pET23a(+) CAG (SEQ ID NO: 50) vector for recombinant protein SbOMT4-RE-R
 CCCAAGCTTCTTTGTGAATTCAAGGG expression in *E. coli* CCC (SEQ ID NO:
 51) SbSOMT-HF-F TGAGAACCTGTACTTCCAAGGCAGC HiFi Assembly of SbSOMT
 (2-377) in TACGACAGCAGCAGTAG (SEQ ID pET-N-His-TEV vector for
 recombinant NO: 52) SbSOMT-HF-R AGCCGGATCTCACTCGAGTTACTTTG protein
 expression in *E. coli* TGAAGTCAAGGGCCCAGAC (SEQ ID NO: 53) SbCOMT-HF-
 F TGAGAACCTGTACTTCCAAGGGTCG HiFi Assembly of SbCOMT (2-362) in
 ACGGCGGAGGAC (SEQ ID NO: 54) pET-N-His-TEV vector for recombinant
 SbCOMT-HF-R AGCCGGATCTCACTCGAGTTACTTGA protein expression in *E. coli*
 TGAAGTTCGATGGCCCAG (SEQ ID NO: 55) SsSTS-Gib-F
 CTTCTGCAGGAATTCGATATCATGAC Gibson Assembly of SsSTS in pET-N-
 TGGGAAGGTAACATTGGGG (SEQ ID His-TEV vector for recombinant protein
 NO: 56) SsSTS-Gib-R GAGAGATCTGTCTGACGATATCCTAC expression in *E. coli*
 ACTGTGATGATGGGAACGC (SEQ ID NO: 57) SsCOMT-Gib-F
 CTTCTGCAGGAATTCGATATCATGGG Gibson Assembly of SsCOMT in pET-
 CTCGACCGCC (SEQ ID NO: 58) N-His-TEV vector for recombinant SsCOMT-Gib-
 R GAGAGATCTGTCTGACGATATCTTACT protein expression in *E. coli*
 TGATGAAGTTCGATGGCCCAG (SEQ ID NO: 59) SbSOMT-1144N-F
 GTGCTTCCGCTTGGGATGATGAACCT Generation of SbSOMT 1144N and
 AAACAAGACATTCCTGGACAGC (SEQ ID NO: 60) SbSOMT-1144N-R
 CATCATCCCAAGCGGAAGCAC (SEQ ID I144N/F337N mutant proteins ID NO: 61)
 SbSOMT-F337N-F CCTCACCATGCTGGTCACGAACGGC Generation of SbSOMT
 F337N and AGTGGTAAAGAGAGGACACA (SEQ ID NO: 62) SbSOMT-F337N-R
 CGTGACCAGCATGGTGAGG (SEQ ID I144N/F337N mutant proteins NO: 63)
 SbSOMT-H282A-F TTCTGCTCAAGTGGATTCTTGCTGAT Generation of SbSOMT
 H282A mutant TGGGACGACAAGGCGTG (SEQ ID NO: 64) SbSOMT-H282A-R
 AAGAATCCACTTGAGCAGAACTGCA protein (SEQ ID NO: 65) SbSOMT-H282N-F
 GATTGGGACGACAAGGCGTG (SEQ ID Generation of SbSOMT H282N mutant ID
 NO: 66) protein SbSOMT-H282N-R GTTAAGAATCCACTTGAGCAGAAC Generation of
 SbSOMT H282N and (SEQ ID NO: 67) H282N/D283A mutant proteins SbSOMT-
 D283A-F GCGTGGGACGACAAGG (SEQ ID Generation of SbSOMT D283A and
 NO: 68) H282N/D283A mutant proteins SbSOMT-D283A-R
 ATGAAGAATCCACTTGAGCAGAAC Generation of SbSOMT D283A mutant (SEQ
 ID NO: 69) protein SbSOMT-E310A-F GCGTACGTTGTTCCGGATG (SEQ ID
 Generation of SbSOMT E310A mutant NO: 70) SbSOMT-E310A-R
 CAGAACGATCACCTTGCCTC (SEQ ID protein NO: 71) SbSOMT-E342A-F
 GCGAGGACACAGAGGGAGTTC (SEQ ID Generation of SbSOMT E342A mutant ID
 NO: 72) SbSOMT-E342A-R TTTACCACTGCCAAACGTGAC (SEQ ID protein ID NO: 73)
 sbsomt-gPCR-F1 TGAATCATCAGCACGGAACG (SEQ ID Genotyping of sbsomt
 CRISPR/Cas9 NO: 74) sbsomt-gPCR-R1 CACCCTCCTAGTAGCCGCAA (SEQ ID mutants
 NO: 75) sbsomt-gPCR-F2 GGCAAAACATTACGGATGCAGT (SEQ ID Genotyping of

sbsomt CRISPR/Cas9 ID NO: 76) sbsomt-gPCR-R2 TGACTTCACGGAGGCTCATAG (SEQ mutants ID NO: 77) SbSTS-PEAQ-F ATCGGACCGGTATGACGACTGGGAA Cloning of SbSTS in pEAQ-HT vector GGTAAC (SEQ ID NO: 78) SbSTS-PEAQ-R CGATCCTCGAGTCATGCAGCCACTGT for *N. benthamiana* expression GGTGA (SEQ ID NO: 79) SbSOMT-PEAQ-F TGCCCAAATTCGCGAATGGGCAGCT Cloning of SbSOMT in pEAQ-HT ACGACAGC (SEQ ID NO: 80) SbSOMT-PEAQ-R TGGTGATGGTGATGCATATTATTTT vector for *N. benthamiana* expression CAAATTGAGGATGAGACCACTTTGT GAACTCAAGGGCCCAG (SEQ ID NO: 81)

TABLE-US-00012 TABLE 12 Protein and ligand concentrations used for ITC setup. Protein concentration Ligand concentration Protein in cell (μM) Ligand in syringe (μM) SbSOMT, WT 50 Resveratrol 750 50 Pinostilbene 750 25 Pterostilbene 500 50 Piceatannol 1000 50 SAM 2000 SbCOMT, WT 50 Resveratrol 750 50 Pinostilbene 750 25 Pterostilbene 500 50 Piceatannol 750 50 SAM 1500 SbSOMT mutants: I144N 25 Resveratrol 1000 D283A 50 Resveratrol 1000 50 Resveratrol 1000 E310A 50 Pinostilbene 1000 25 Pterostilbene 500 F337N 25 Resveratrol 1000 E342A 50 Resveratrol 1000 I144N/F337N 25 Resveratrol 1000 H282N/D283A 25 Resveratrol 1000 SsCOMT, WT 50 Resveratrol 750 50 Pinostilbene 750 25 Pterostilbene 500 50 Piceatannol 1000 50 SAM 2000

TABLE-US-00013 TABLE 13 X-ray data collection and refinement statistics of disclosed structures. SbSOMT- SbSOMT- SbSOMT- Resveratrol- Pinostilbene- Pterostilbene- SbSOMT- β-NAD β-NAD β-NAD Resveratrol Data collection Wavelength (Å) 0.9785 0.9784 0.9784 0.9785 Resolution (Å) .sup.a 48.42-1.72 (1.76-1.72) 19.70-2.10 (2.14-2.10) 31.40-2.40 (2.44-2.40) 48.91-2.56 (2.65-2.56) Space group P 3.sub.1 2 1 P 3.sub.1 2 1 P 3.sub.1 2 1 P 2.sub.1 2.sub.1 2.sub.1 Cell dimensions a, b, c (Å) 96.8, 96.8, 168.0 96.7, 96.7, 166.9 96.5, 96.5, 169.4 97.5, 111.7, 131.1 α, β, γ (°) 90, 90, 120 90, 90, 120 90, 90, 120 90, 90, 90 Total reflections 1882940 651200 709365 616499 Unique 97225 101522 36363 46864 reflections Multiplicity 19.4 6.4 19.5 13.2 Completeness 99.8 (97.1) 99.8 (99.0) 99.2 (99.9) 99.9 (100.0) (%) .sup.a Mean I/sigma (I) .sup.a 30.2 (2.9) 9.2 (2.0) 7.5 (0.8) 16.3 (2.4) Wilson B-factor 22.6 31.43 36.78 52.2 (Å.sup.2) R.sub.merge (%) .sup.a 6.0 (94.1) 21.8 (72.4) 19.9 (95.5) 11.8 (115.6) R.sub.meas (%) .sup.a 6.3 (101.0) 23.8 (78.8) 20.4 (98.2) 12.7 (124.7) R.sub.pim (%) .sup.a 2.0 (35.7) 9.4 (31.0) 4.6 (22.8) 4.8 (46.8) CC.sub.1/2 .sup.a 1.000 (0.849) 0.891 (0.790) 0.915 (0.871) 0.999 (0.902) .sup.a Value relative to the highest resolution shell are given in parentheses.

TABLE-US-00014 TABLE 13 X-ray data collection and refinement statistics of structures reported in this study (continued). SbSOMT- SbSOMT- SbSOMT- Resveratrol- Pinostilbene- Pterostilbene- SbSOMT- β-NAD β-NAD β-NAD Resveratrol Refinement Reflections used in 97159 53133 36012 46802 refinement Reflections used for R- 4899 2652 1899 2368 free R-work 0.163 0.171 0.187 0.182 R-free 0.190 0.202 0.246 0.252 Number of atoms 6638 6216 6104 11441 Macromolecules 5841 5674 5721 11286 Ligands/Ions 229 274 214 92 Solvent 568 268 169 63 rmsd .sup.b bond lengths 0.012 0.015 0.016 0.016 (Å) rmsd .sup.b bond angles (°) 1.73 2.023 2.10 2.22 Ramachandran plot Favored (%) 99.03 97.36 95.57 95.72 Allowed (%) 0.97 2.22 3.60 3.31 Outlier (%) 0.00 0.42 0.83 0.97 Sidechain outlier (%) 0.64 2.64 1.96 4.56 Molprobit Clash score 4.46 8.69 7.95 8.10 Mean B-value (Å.sup.2) 29.0 37.0 46.0 68.0 Macromolecules 28.38 36.72 45.54 62.15 Ligands/Ions .sup.c 53.31 (41.01) 61.19 78.06 52.78 Solvent 36.14 39.74 43.73 41.98 PDB ID code 7VB8 7WAR 7WAS 7WAQ .sup.b rmsd, root mean square deviation. .sup.c The mean B-values (Å.sup.2) of ligands and ions are given separately, with value for ion given in parentheses.

TABLE-US-00015 TABLE 14 Accession numbers of sequences used in this study. Protein GenBank accession number SEQ ID NO *Arabidopsis thaliana* AtOMT1 NP_200227 5 *Carthamus tinctorius* CtCaldOMT BAG71895 6 *Catharanthus roseus* CrCOMT Q8W013 7 *Hordeum vulgare* HvCOMT ABQ58825 8 *Medicago sativa* MsCOMT P28002 9 *Medicago sativa* MsIOMT AAC49927 10 *Oryza sativa* OsCaldOMT1 Q6ZD89 11 *Oryza sativa* OsNOMT QOIP69 12 *Panicum virgatum* PvCOMT ADX98508 13 *Pinus sylvestris* PsPMT2 AQX17823 14 *Pinus taeda*

PtAOMT Q43096 15 *Populus tremuloides* PtCOMT AAB61731 16 *Rosa hybrid* RhOOMT AAM23004 17 *Solanum lycopersicum* SlCOMT XP_004235028 18 *Sorghum bicolor* SbCOMT AAO43609 2 *Sorghum bicolor* SbNOMT XP_002465645 19 *Sorghum bicolor* SbOMT1 ABP01563 20 *Sorghum bicolor* SbOMT3 ABP01564 21 *Sorghum bicolor* SbOMT4 XP_002443937 22 *Sorghum bicolor* SbSOMT XP_021320201 1 *Sorghum bicolor* SbSTS1 XP_002445139 23 *Triticum aestivum* TaCOMT1 AAP23942 24 *Triticum aestivum* TaCOMT2 Q38J50 25 *Vitis vinifera* VvCOMT NP_001268100 26 *Vitis vinifera* VvROMT CAQ76879 27 *Zea mays* ZmCOMT Q06509 28 *Zea mays* ZmNOMT XP_020399799 29 Gene Ensembl Plants Identifier *Saccharum spontaneum* SsSTS Sspon.06G0010290-2P 3 *Saccharum spontaneum* SsCOMT Sspon.06G0010980-3C 4 Gene GenBank accession number *Sorghum halepense* ShOMT1 GGDZ01007645 30 *Sorghum halepense* ShSOMT GGDZ01140964 31

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TABLE-US-00016 Additional Sequences AtOMT1 (*Arabidopsis thaliana*; NP_200227) (SEQ ID NO: 5) MGSTAETQLT PVQVTDDEAA LFAMQLASAS VLPMAKLSAL ELDLLEIMAK NGSPMSPTTEI ASKLPTKNPE APVMLDRILR LLTSYSVLTC SNRKLSGDGV ERIYGLGPVC KYLTKNEDGV SIAALCLMNQ DKVLMESWYH LKDAILDGGI PFNKAYGMSA FEYHGTDPFR NKVENNGMSN HSTITMKKIL ETYKGFEGLT SLVDVGGGIG ATLKMIVSKY PNLKGINFDL PHVIEDAPSH PGIEHVGGDM FVSVPKGDAI FMKWICHDS DEHCVKFLKN CYESLPEDGK VILAECILPE TPDSSLSTKQ VVHVDCIMLA HNPGGKERTE KEFEALAKAS GFKGIKVVCD AFGVNLIELL KKL CtCaldOMT (*Carthamus tinctorius*; BAG71895) (SEQ ID NO: 6) MAHLEEEEF LFAMQLASAS VLPMLVLSAI DLDLLEIIAK AGPGAHVSPA YLAAQLPKAD NPEAAVMLDR ICRLATYSV LTCTLKLDH GDGDVERLYG LAPVCKFLVK NEDGVSNAPL LLMNQDKILM ESWYHLKDAV LDGGIPFNKA YGMSAFEYHG KDERFNKVFN SGMFNHSTMT MKKILDVYPG FNGVKTLVDV GGGTGASLSM ILSKHPSIKG INFDLPHVIQ DATNYPGIEH VGGDMFESVP KGDAIFMKWI CHDWSDAHCL KFLKNKYKAL PENGKVIVAE CILPETPDSS LATKNVVHID VIMLAHNPGG KERTEKEFQA LAKAAGFKGF HKPCSALNTW LMEFTK CrCOMT (*Catharanthus roseus*; Q8W013) (SEQ ID NO: 7) MGSANPDNKN SMTKEEEFAC LSAMRLASAS VLPMLVLSAI ELDLLELIKK SGPGAYVSPS ELAAQLPTQN PDAPVMLDRI LRLASYSVL NCTLKDLPDG

GIERLYSLAP VCKFTLTALKED GVSMALLESW NQDKVLMESW YHLKDAVLEG
 GIPFNKAYGM TAFEYHGKDP RENKVFNQGM SNHSTIIMKK ILEIYQGFQG
 LKTVDVGGG TGATLNMIVS KYPISGINF DLPHVIEDAP SYPGVDHVGG
 DMFVSVPKGD AIFMKWICH D WSDAHCLKFL KNCHEALPEN GKVILAECLL
 PEAPDSTLST QNTVHVDVIM LAHNPPGGKER TEKEFEALAK GAGFRGFIKV
 CCAYNSWIME LLK HvCOMT (*Hordeum vulgare*; ABQ58825)(SEQ ID NO: 8)
 MGSIAAGADE DACMYALQLV SSSILPMTLK NAIELGLLET LMSAGGKFLT
 PAEVAACKLPS TANPEAPDMV DRMLRLLASY KVVSCRTEES KDGRLSRRYG
 AAPVCKYLT NEDGVSMSAL ALMNQDKVLM ESWYYLKDAV LDGGIPFNKA
 YGMSAFEYHG TDPRFNRVFN EGMKNHSIII TKKLLESYKG FEGLGTLVDV
 GGGVGATVGA IIARYPAVKG INFDLPHVIS EAPAFPGVTH IGGDMFQKVP
 SGDAILMKWI LHDWSDEHCA TLLKNCYDAL PAHGKVVLVE CILPVNPEAT
 PEVQGVFHV D MIMLAHNPPG RERYEREFEA LAKGAGFAAM KTTYIYANAW
 AIEFTK MsCOMT (*Medicago sativa*; P28002)(SEQ ID NO: 9) MGSTGETQIT
 PTHISDEEAN LFAMQLASAS VLPMLKSAL ELDLLEIIAK AGPGAQISPI
 EIASQLPTTN PDAPVMLDRM LRLACYIIL TCSVRTQQDG KVQRLYGLAT
 VAKYLVKNED GVSISALNLM NQDKVLMESW YHLKDAVLDG GIPFNKAYGM
 TAFEYHGTDP RFNKVENKGM SDHSTITMCK ILETYTGFEG LKSLVDVGGG
 TGAVINTIVS KYPTIKGINF DLPHVIEDAP SYPGVEHVGG DMFVSIPKAD
 AVFMKWICH D WSEHCLKFL KNCYEALPDN GKVIVAECIL PVAPDSSLAT
 KGVVHIDVIM LAHNPPGGKER TQKEFEDLAK GAGFQGFKVH CNAFNTYIME
 FLKKV MsIOMT (*Medicago sativa*; AAC49927)(SEQ ID NO: 10) MASSINGRKP
 SEIFKAQALL YKHIYAFIDS MSLKWAVGMN IPNIIHNHGK PISLSNLVSI
 LQVPSSKIGN VRRLMRYLAH NGFFEITKE EESYALTVAS ELLVRGSDLC
 LAPMVECVLD PTLSGSYHEL KKWIYEEDLT LFGVTLGSGF WDFLDKNPEY
 NTSFNDAMAS DSKLINLALR DCDFVEDGLE SIVDVGGGTG TTAKIICETF
 PKLKCIVEDR PQVVENLSGS NNLTYVGGDM FTSIPNADAV LLKYILHNWT
 DKDCLRILKK CKEAVTNDGK RGKVTHIDMV INEKKDENQV TQIKLLMDVN
 MACLNGKERN EEEWKKLFIE AGFQHYKISP LTGFLSLIEI YP OsCAlDOMT1
 (*Oryza sativa*; Q6ZD89)(SEQ ID NO: 11) MGSTAADMAA AADEEACMYA
 LQLASSILP MTLKNAIELG LLETQLSAAV AGGGGKAALL TPAEVADKLP
 SKANPAAADM VDRMLRLLAS YNVVRCMEE GADGKLSRRY AAPVCKWLT
 PNEDGVSMMA LALMNQDKVL MESWYYLKDA VLDGGIPFNK AYGMTAFEYH
 GTDARFNRVF NEGMKNHSVIITKKLLDLYT GFDAASTVVD VGGGVGATVA
 AVVSRHPHIR GINYDLPHVISEAPPFGVE HVGGDMFASV PRGGDAILMK
 WILHDWSDEH CARLLKNCYDALPEHGKVVV VECVLPESD ATAREQGVFH
 VDMIMLAHNP G GKERYEREFE RELARAAGFT GFKATYIYAN AWAIEFTK
 OsNOMT (*Oryza sativa*; QOIP69)(SEQ ID NO: 12) MVSPVVHRHA
 AGGGSGGDDD DQACMYALEL LGGSVVSMTL KAAIELGLVD ELLAAAGAAV
 TAEELAAARLR LPAVAAAAA VDRMLRLLAS YGVVRCATEA GPDGKARRSY
 AAPVCKWLA AGSSSGEGSM APLGLLNLDK VFMENWYYLK EAVSEG GTAF
 DKAYGTS LFQ YLGQDGNEPS NTLFNQAMAS HSVVITNKLL QFFRFGFDAGA
 GVDVLVDVGG GVGATLRMIT ARHPHLRGVN YDLPHVIAQA PPVEGVEHIG
 GSMFDHVPSG SAILLKWILH LWGDEECVKI LKNCYKALPA KGKVILVEYV
 LPASPEATLA AQEAFRLDVM MLNRLAGGKE RTQQEFTDLA VDAGFSGDCK
 PTYIFTNVWA LEFTK PvCOMT (*Panicum virgatum*; ADX98508)(SEQ ID NO:
 13) MGSTATDVAA AADEEACMYA LQLASSILP MTLKNAIELG LLEV LQKDPA
 AALAPEEVVA QLPVAPANPD AAAMVDRMLR LLASYDVVRC QMEEGK DGRY
 SRRYAAPVC KWLTPNEDGV SMAALALMNQ DKVLMESWYY LKDAVLEGGI
 PFNKAYGMTA FEYHGTDP RF NRVFNEGMKN HSVIITKKLL E FYAGFEGVG

TLVDVGGVGVG ATLTAHDSRY PGHVEADPPF PGHVEADPPF PGHVEADPPF
 FKAHPAGDAI LMKWILHDWS DAHCAAILKN CYDALPAGGK VIAVECILPV
 NPEATPKAQG VFHVDMMILA HNPGGKERYE REFEELAKGA GFTGFKATYI
 YANAWAIEFT K PsPMT2 (*Pinus sylvestris*; AQX17823)(SEQ ID NO: 14)
 MNMQSVKDEE ALRASALGLA FSLETPFLLK CAIRLKIPDI ISKAGPDVSL
 SVHQIAAQLP SEDPDMGALS RILTYLSTMG ILQAIVPPEG VNAPMNIRYG
 LTNLTKTYFT SEDISSRSLV PFVLLQTHPL YVTAWDNIHE RVLHGGDNFK
 NSSGNGKDFW NFAAGEPEFN AIFNAGMVSU TKATITYVLA VYDGFKDINT
 LVDVGGGRGE ALSLITEAHP HIRAINFDLP QVIATAPTIP GVQHMSGNLF
 ESAPSADAIF MKNFLHSWND EDCIKLLNNC HQALPEKGL ILSEAILDLT
 EGSDMIGSAN VLDVAVMLNCL PGGGERTRKQ WNDLLQAAGF SISKIVGRNG
 TLTKVIEAIK S PtAEOMT (*Pinus taeda*; Q43096)(SEQ ID NO: 15)
 MDSNMNGLAK SNGCEISRDL FFESEEEELQ GQAEAWKCTF AFAESLAVKC
 VVLLGIPDMI AREGPRATLS LCEIVANLPT ESPDAACLFR IMRFLVAKGI
 FPASKSARRR AFETRYGLTP ASKWLVKGRE LSMAPMLLMQ NDETTLAPWH
 HFNECVLEGG VAFQKANGAE IWSYASDHPD FNNLFNNAMA CNARIVMKAI
 LSKYQGFHSL NSLVDVGGGT GTAVAEIVRA YPFIRGINYD LPHVVATASS
 LSGVQHVGGD MFETVPTADA IFMKWIMHDW NDEDCIKILK NCRKAIPDTG
 KVHVDVVDL ADQGDNTDKK RKKAVDPVIG TVFDLVMVAH SSGGKERTEK
 EWKRILLEGG FSRYNHIEP ALQSVIEAFP R PtCOMT (*Populus tremuloides*;
 AAB61731)(SEQ ID NO: 16)MGSTGETQMT PTQVSDEEAH LFAMQLASAS
 VLPMLKTAI ELDLLEIMAK AGPGAFLSTS EIASHLPTKN PDAPVMLDRI
 LRLLASYSIL TCSLKDLPDG KVERLYGLAP VCKFLTKNED GVSVSPLCLM
 NQDKVLMESW YYLKDAILDG GIPFNKAYGM TAFEYHGTDP RFNKVFNKGM
 SDHSTITMKK ILETYKGFEG LTSLVGVGGG TGAVVNTIVS KYPISKGINF
 DLPHVIEDAP SYPGVEHVGG DMFVSVPKAD AVFMKWICH D WSDAHCLKFL
 KNCYDALPEN GKVLVECIL PVAPDTSLAT KGVVHVDVIM LAHNPGGKER
 TEKEFEGLAK GAGFQGFVEM CCAFNTHVIE FRKKA RhOOMT (*Rosa hybrid*;
 AAM23004)(SEQ ID NO: 17)MERLNSFRHL NQKWSNGEHS NELLHAQAH
 WNHIFSINS MSLKSAIQLG IPDIINKHGY PMTLSELSTA LPHPTKSHS
 VYRLMRILVH SGFFAKKKLS KTDEEGYTTL DASQLLLKDH PLSLTPYLTA
 MLDPVLTNPW NYLSTWFQND DPTPFDTAHG MTFWDYGNHQ PSIAHLEND
 MASDARLVTS VIINDCKGVF EGLESVLDVG GGTGTLAKAI ADAFPHIECT
 VLDLPHVVAD LQGSKNLKYT GGDMFEAVPP ADTVLLKWIL HDWSDEECIK
 ILERSRVAIT GKEKKGKVII IDMMMENQKG DEESIETQLF FDMMLMALVG
 GKERNEKEWA KLFTDAGFSD YKITPISGLR SLIEVYP SICOMT (*Solanum
 lycopersicum*; XP_004235028)(SEQ ID NO: 18)MGSTSLTQTE DEAFLFAMQL
 ASASVLPMLV KSAVELELLE LMAKAGPGAS ISPAELASQL PCKNPDPAPVM
 LDRMLRLLAA YSVLNCTLR TLPDGRVERLY SLAPVCKFLT KNADGVSVAP
 LLLMNQDKVL MQSWYHLKDA VLDGGIPFNK AYGMTAFEYH GTDPRFNKVF
 NRGMSDHTTL SIKKILEDYK GFEGLSIVD VGGGTGATVS MIVSKYPSIK
 GINFDLPHVI EDAPAYPGVE HIGGDMFVSV PKADAIFMKW ICHDWSDEHC
 LKFLKNCYEA VPANGKVIIA ECLLPEVPDT SSSTKNTVHV DVIMLAHNPG
 GKERTEKEFE ALAKGAGENG FTKASCAYNT WIMEFTK SbNOMT (*Sorghum
 bicolor*; XP_002465645)(SEQ ID NO: 19)MACTTAASLQ HDKANDDEAC
 MYAQELLFSF IVPMTLKAVI ELGLIDYLLAADGRSVTPPE LAAEWPQSAE
 AAAAVDRMMR LLASHSVVRC TTEVGPDGKARRSYAAPVC KWLATRNAGG
 QGSLAPMGLM NLNKAFMETW YFMKEAVTEG ATPTEKAYGM PLFEHLGSDE
 ASNTLFNQAM AGHSEMIKK LLEVYRGFEG VDLVLDVGGG TGSTLRMVTA
 QYKHLRGVNY DLPHVIAQAP PVQGVHVGG SMFEYIPSGN AILLKWILHL

WRDCEVKIL KNCRRALPAN GKSHVEYVL PASPEPTQVA QVSLLDLVAM
 LNRLRGAKER TEQEFAQLAA EAGFSGGCRA TYVFASAWAL EFTK SbOMT1
 (*Sorghum bicolor*; ABP01563)(SEQ ID NO: 20) MASYTSTSGQ FAVGKVAAAN
 QDDETCMHAL KLLGGLAVPF TIKAVIELGI MDLLAADRA MTAEALTAAL
 LCPAPAPAAA AAMVDRMLRF LASHGVVRCA TESEELGSDD GKSCRRYAAA
 PVCKWFARGG GVESVPMGF WMTSTTNMET WHNIKDGVLA GETPFDKAYG
 MPVFEYLGAN GTMNTLFNEA MASHSMITK RLLEVFRGFE NYSVLVDVGG
 GNGTTMQMIR SQYENISGIN YDLPHVIAQA SPIEGVEHVA GNMEDNIPRG
 DAIILKWILH NWGDKECVKI LKNCYTALPV NGTVIILEYI LPETPEETLA
 SQLAFDFDLG MMLFFGASGK ERTEKELLEL AREAGFSGDY TATYIFANVW
 AHEFTK SbOMT3 (*Sorghum bicolor*; ABP01564)(SEQ ID NO: 21)
 MVLISEDRE LLQAHVELWN QTYSEFMKSA LAVALDLHIA DAIHRRGGAA
 TLSQILGEIG VRPCKLPGLH RIMRVLTVSG TFTIVQPSAE TMSSESDGRE
 PVYKLTTASS LLVSSESSAT ASLSPMLNHV LSPFRDSPLS MGLTAWFRHD
 EDEQAPGMCP FTLMYGTTLW EVCRRDDAIN ALFNNAMAAD SNFLMQILLK
 EFSEVFLGID SLVDVAGGVG GATMAIAAAF PCLKCTVLDL PHVVAKAPSS
 SIGNVQFVGG DMFESIPPAN VVLLKWILHD WSNDECILK KNCKQAIPSR
 DAGGKIIID VVVGSDSSDT KLETTQVIYD LHLMKIGGVE RDEQEWKKIF
 LEAGFKDYKI MPILGLRSII ELYP SbOMT4 (*Sorghum bicolor*; XP_002443937)
 (SEQ ID NO: 22) MGSYTTRVHA VEKAANNQDD ETCMHALLMLL
 GGLAVPCTIK AVIELGIMDL LLAADRAMTA EELTARLPCP AAATAAAMVD
 RMLRFLASHG VVRCAAATKE SSELGSDGGK SCRRYAAAPV CRWFTRSGGV
 ESVVPMGLWM TGKTVLETWY NIKEAVLEGE TPFDRAYGQP FFEYLGNGT
 VNTLFDEAMA NHSTIITKRL VEVFRGFENY SVLVDVGGNK GTTLQMIRSQ
 YENISGINYD LPRVIAQAPP IEGVEHVGGN MFDNVPRGDA IILKWILHDW
 GDKDCVKILK NCYAALPVNG TMIILEYILP ETPEETLTSQ LAFNFDFGMM
 LMYGAKGKER TEKELSELAR EAGFSGDCTA TYIFASI WAL EFTK SbSTS1
 (*Sorghum bicolor*; XP_002445139)(SEQ ID NO: 23) MTTGKVTLA
 VRKAQRAEGP ATVLAIGTAT PANCVYQADY PDYYFRVTKS EHLTDLKEKF
 KRICHKSMIR KRYMHLTEI LEENPNMSSY WAPSLDARQD ILIQEIPKLG
 AEAAEKALKE WGQPRSRITH LVFCTTSGVD MPGADYQLIK LLGLCPSVNR
 AMMYHQGCFA GGMVLRRLAKD LAENNRGARV LIVCSEITVV TFRGPSESHL
 DSLVGQALFG DGAAAVIVGA DPSEPAERPL FHLVSASQTI LPDSEGAIEG
 HLREVGLTFH LODRVPQLIS MNIERLLEDA FAPLGISDWN SIFWVAHPGG
 PAILNMVEAK VGLDKARMCA TRHILAEYGN MSSVCVLFIL DEMRNRSAKD
 GHTTTGEGME WGVLFSGFPG LTVETIVLHS VPITTVA TaCOMT1 (*Triticum
aestivum*; AAP23942)(SEQ ID NO: 24) MGSTAADMAA SADEEACMYA
 LQLVSSSILP MTLKNAIELG LLETLVAAGG KLLTPAEVAA KLPSTANPAA
 ADMVDRMLRL LASYNVVSCT MEEGKDGRS RRYRAAPVCK FLTPNEDGVS
 MAALALMNQD KVLMEWYYL KDAVLDGGIP FNKAYGMSAF EYHGTDPRFN
 RVFNEGMKNH SIITKKLLE VYKGFEGLT IVDVGGGVGA TVGAITAAYP
 AIKGINFDLP HVISEAQFP GVTHVGGDMF QKVPSGDAIL MKWILHDWSD
 EHCATLLKNC YDALPAHGKV VLVECILPVN PEATPKAQGV FHVDMIMLAH
 NPGGRERYER EFEALAKGAG FKAIKTTYIY ANAFAIEFTK TaCOMT2 (*Triticum
aestivum*; Q38J50)(SEQ ID NO: 25) MGSIAAGADE DACMYALQLV
 SSSILPMTLK NAIELGLLET LMAAGGKFLT PAEVA AKLPS AANPEAPDMV
 DRMLRLLASY NVVSCRTEG KDGRSRRYG AAPVCKYLT NEDGVSMSAL
 ALMNQDKVLM ESWYYLKDAV LDGGIPFNKA YGMSAFEYHG TDPRFN RVFN
 EGMKNHSIII TKKLLESYKG FEGLGTLVDV GGGVGATVAA ITAHYPTIKG
 INFDLPHVIS EAPPFPGVTH VGGDMFQKVP SGDAILMKWI LHDWSDEHCA

TLLKNVCYDAL PAHGKYNVLVE CILKVNPEAT PKAQGVFHV D MIMLAHNPGG
 RERYEREFEA LAKGAGFAAM KTTYIYANAW AIEFTK VvCOMT (*Vitis vinifera*;
 NP_001268100)(SEQ ID NO: 26) MESTLAFNSG SNSMNQSFSS SAEFNSPVPE
 TIPKSEEDTF VFATLLTSAS VLPMALKSAL ELDLLEIIAK AGPGA FVSTS
 EIAAKITKRN PKAPVMLDRI LRLLATYDVV KCSLRDSPDG GVERLYGLGP
 VCKYFTTNE D GVSVA PLLLM NQDKVPMQSK RYHLKDAVLD GGIPFNKAYG
 MTD FEYHGTE PRENKVENNG VSGHPTITMK KILEAYKGFE GLTSIVDVGG
 GTGATLNMII SKYPTIKGIN FDLPHVIDDA PSYPGVEHVG GDMFVSVPKG
 DAIFMKWMCY EWDDAHCLKF LENCYQALPD NGKVIVAECI LPVVPDTS LA
 TKS AVHIDVI MLAYNTGGKARTEKEFEALA KGAGFQGFKV VCCAFNSWIM
 EFCKTA VvROMT (*Vitis vinifera*; CAQ76879)(SEQ ID NO: 27) MDLANGVISA
 ELLHAQAHVW NHIFNFIKSM SLKCAIQLGI PDIIHNHGKP MTLPELVAKL
 PVHPKRSQCV YRLMRILVHS GFLAAQRVQQ GKEEEGYVLT DASRLLLMDD
 SLSIRPLVLA MLDPI LTKPW HYLSAWFQND DPTPFHTAHE RSFWDYAGHE
 PQLNNSFNEA MASDARLLTS VLLKEGQGVF AGLNSLVDVG GGTGKVAKAI
 ANAFPHLNCT VLDLPHVVAG LQGSKNLNYF AGDMFEAIPP ADAILLKWIL
 HDWSDEECVK ILKRCREAI P SKENGGKVII IDMIMMKNQG DYKSTETQLF
 FDMTMMIFAP GRERDENEWE KLFLDAGFSH YKITPILGLR SLIEVYP ZmCOMT
 (*Zea mays*; Q06509)(SEQ ID NO: 28) MGSTAGDVAA VVDEEACMYA
 MQLASSSILP MTLKNAIELG LLEV LQKEAG GKKAALAPEE VVARMPAAPG
 DPAAAAAMVD RMLRLLASYD VVRCQMEDRD GRYERRY SAA PVCKWLTPNE
 DGVSMAALAL MNQDKVLMES WYYLKDAVLD GGIPFNKAYG MTAF EYHGTD
 SRFNRVFN EG MKNH SVIITK KLLDFYTGF E GVSTLV DVGG GVGATLHAIT
 SRHPHISGVN FDLPHVISEA PPFGVRHVG GDMFASVPAG DAILMKWILH
 DWSDAHCATL LKNCYDALPE NGKVIVVECV LPVNTEATPK AQGVFHV DMI
 MLAHNP GGKE RYEREFRELA KGAGFSGFKA TYIYANAWAI EFIK ZmNOMT
 (*Zea mays*; XP_020399799)(SEQ ID NO: 29) MACTTAASLQ HDRANDDEAC
 MYAQELLSCF VVPMTLKAVI ELGLIDDLLA ADGRFVTPEE LAARWARPAE
 AAAAVDRMLR FLASHSVVRC TTEAAGPDGR ARRSYAAAPV CKWLIARNGT
 GQGSWAPIGL MNLNKGFMET WYYMKDAVAE GATPTEKAYG MPLFEHLGSD
 EALNTLFNQA MAGHSEIVIS KLL EYVRGFE GVDVLVDVGG GTGSTLRMVT
 AQYKHLRGVN YDLPHVIAQA PPVQGVEHAG GSMFEYIPSG NAILLKWILH
 LWRDEECIKI LKNCHRALPA NGKVIVVECV LPASPEPTQV AQGALLLDVV
 MLNRLRGAKE RTEREFTELA AEAGFSGGCR ATYVFTGAWA LEFTK ShOMTI
 (*Sorghum halepense*; GGDZ01007645)(SEQ ID NO: 30) atgggcagct atactaccag
 agtccacgcc gtagagaaag cagctaaca ccaagatgat gagacgtgca tgc atgcttt gatgctcctt
 ggcggattgg ccgtaccttg taccatcaag geggtcateg agctcggcat catggacctt ctctcgceg
 cggacegcgc catgaccgeg gaggagctca cggcaaggtt gccatgtcct gctgctgcta ctgctgctgc
 tatggctgcac cgc atgctcc gettcctage ttcgcacggc gtggctcaggt gegccgccgc caccaaggag
 tcgtcggagc tgggctccga tgg tggcaag agctgccggc gctacgcggc ggcgccggtg tgcaggtggt
 ttacaaggag cggcggcgctg gagtcgggtg ttccaatggg gttgtggatg accggcaaga ccgtcttgga
 gacctggtac aacataaaaag aagcagtggt agagggagaa acaccattg acagagcata cggccagcca
 tttttgagt accttggtga aaacgggaca gtgaacacgt tgttcgatga ggcaatggcg aaccattcga
 caatcataac gaagaggctg gtcgaggtct tccgtggctt tgagaattac agtgtgctcg tcgatgtcgg
 cggcaacaaa ggcaccacgt tgcaaatgat tagaagtcag tatgagaata ttagtggcat caactacgac
 ctctctctg taattgcgca ggcacctcca attgaagggt tggagcatgt tgg tggcaac atgtttgata
 atgttcacg tggagatgca attatcctaa agtggattct tcacgattgg ggcgacaagg actgcgtcaa
 gatcctaaag aattgctatg cagctctccc ggtgaacggg acgatgatca ttctggagta catcctccca
 gagacaccag aaagaaacact aacategcaa ctagcgttca acttcgattt cgggatgatg ctcatgtacg
 gogccaaagg taaggagaga acagagaaa agttgtcgga gctcgccaga gaggccggt tctctggaga

ctgcacggct acatacatat tegccagcat ctgggccctt gaattcacaa agtaa ShSOMT (*Sorghum*
halepense; GGDZ01140964) (SEQ ID NO: 31) tcgtcctcta tctgtggcca ccaactgctg
tgctatttga gtacacagat acagatgggc agctacgaca gcagcagtag tagtagtaat gactcatcag
cacggaacga ggaggacgag tcgtgcatgt ttgctttgaa gctcctcggc ggggttcgccc tacctttcac
catcaaggcg gtgattgagc tgggcgtcat ggaccagctc ctgactgccg aacgcgcgat gagcgcggag
gagctggtgg cagcagcagt ggcagcacag ctgccacggc ccgaggtagc ctgcaccatg gtggaccgcc
tgctccgttt ccttgccctg cacagcgtag tccggtgcac gaccgaggtg gtggtgggca cggacgacgc
caccaccacc acctgetgcc gccggagcta cgcegcgtca cccgtctgca agtgggtcgc caggaacggc
gtcagaggatt cgggtgcttc gcttgggatg atgaccta acaagacatt cctggacagc tggcaaaaca
ttacggatgc agtgttgaa ggagcagcac catttgagaa aacctacggg atgccaatgt tegagtacct
aagtacaaac ggaccattga acacggtgtt ccacgaggca atggcaaattc attcgatgat tataaccaag
aaactgctca agttcttcg cggcttcgaa ggccttgatg tgctggtcga cgtaggcggc ggcaacggca
ccacgttgca aatgattaga ggtcaatata agaatatgag aggcataaac tacgaccttc ctcatgtcat
tgcgcaggct gcaccagttg aaggtgtgga acatgtaggt ggcagcatgt tcgataatat tccacgegga
aatgcagttc tgctcaagtg gattcttcat gattgggacg acaaggcgtg catcaagatc ctaaagaatt
gctatacagc tctccatgtg agaggcaagg tgatcgttct ggagtacgtt gttccggatg aaccagaacc
tactcttgca gctcaggggtg ccttcgaact ggacctcacc atgctggtca cgtttggcag tggtaaagag
aggacacaga gggagttctc cgagctcgcc atggaggccg gcttctctag agagtttaag gctacgtata
tctttgccaa cgtctgggcc cttgagttca caaagtaa SsCOMT (*Saccharum spontaneum*) (SEQ
ID NO: 32) MGSTAEDVAA VADEEACMYA MQLASASILP MTLKNALELG
LLEVLQAEAP AGKALAPEEV VARLPVAPT N PDAADMVDRM LRLASYDVV
KCQMEDKDGK YERRYSAAPV GKWLTPNEDG VSMAALTLMN QDKVLMESRY
YLKDAVLDDG IPFNKAYGMT AFEYHGTDPR FNRVFNEGMK NHSVIITKKL
LEFYTGFEV STLVDVGGGI GATLHAITSH HPQIKGINFD LPHVISEAPP
FPGVQHVGGD MFKSVPAGDA ILMKWILHDW SDAHCAATLLK NCYDALPENG
KVIVVECVLP VNTEAVPKAQ GVFHVDMMIL AHNPGGRERY EREFHDLAKG
AGFSGFKATY IYANAWAIEF IK SsSTS (*Saccharum spontaneum*) (SEQ ID NO: 33)
MTGKVTLGAV RKAQRAEGSA AVLAIGTATP ANCVYQADYP DYYFRVTKSE
HLTDLKEKFK RICHKSMITK RYMHLTEGFL QENPNMSSYS APSLDARQDI
LIEEVPKLGA AAAEKALKEW GQPRSQITHL VFCTTSGVDM PGADYQLIKL
LGLSLSVNRA MMYHQGCFAG GMVLRLAKDL AENNRGARVL IVCSEITAVT
FRGPSESHLD SLVGQALFGD GAAAVIVGAD PSAAEWPLFQ LVSASQTILP
DSEGAIEGHL REVGLTFHLQ DRVPQLISTN IERLLEDAFT PLGISDWSNI
FWVAHPGGPA ILNMVEAKAG LDKARLCATR HILAEYGNMS SACVLFILDE
MRNKSAEDGH TTTGEGMEWG VLFGFGPGLT VETIVLQSVP IITV ShOMT1
(*Sorghum halepense*) (SEQ ID NO: 34) MGSYTTRVHA VEKAANNQDD
ETCMHALMLL GGLAVPCTIK AVIELGIMDL LLAADRAMTA EELTARLPCP
AAATAAAMVD RMLRFLASHG VVRCAAATKE SSELGSDGGK SCRRYAAAPV
CRWFTRSGGV ESVVPMLGWM TGKTVLETWY NIKEAVLEGE TPFDRAYGQP
FFEYLGNGT VNTLFDEAMA NHSTIITKRL VEVFRGFENY SVLVDVGGNK
GTTLQMIRSQ YENISGINYD LPRVIAQAPP IEGVEHVGGN MFDNVPRGDA
IILKWILHDW GDKDCVKILK NCYAALPVNG TMILEYILP ETPEETLTSQ
LAFNFDFGMM LMYGAKGKER TEKELSELAR EAGFSGDCTA TYIFASIWAL EFTK
ShSOMT (*Sorghum halepense*) (SEQ ID NO: 35) MGSYDSSSSS SNDSSARNEE
DESCMFALKL LGGFAVPFTI KAVIELGVMD QLLTAERAMS AEELVAAAVA
AQLPRPEVAC TMVDRLRLFL ASHSVVRCTT EVVVGTTDDAT TTTCRRSYA
ASPVCKWFA R NGVEDSVLPL GMMILNKTF L DSWQNITDAV LEGAAPFEKT
YGMPPMEYLS TNGPLNTVFH EAMANHSMII TKKLLKFFRG FEGLDVLVDV
GGNGTTLQM IRGQYKNMRG INYDLPHVIA QAAPVEGVEH VGGSMFDNIP
RGNVLLKWI LHDWDDKACI KILKNCYTAL HVRGKVIVLE YVVPDEPEPT

LAAGAFLEF LTMVTFGSG KERTQREFSRE FKATYIFANV
WALEFTK MsIOMT (*Medicago sativa*; P93324) (SEQ ID NO: 82) MGNSYITKED
NQISATSEQT EDSACLSAMV LTTNLVYPV LNAIDLNLFI EIAKATPPG
AFMSPSEIAS KLPASTQHSD LPNRLDRMLR LLASYSVLTS TTRTIEDGGA
ERVYGLSMVG KYLVPDESRG YLASFTTFLC YPALLQVWMN FKEAVVDEDI
DLFKNVHGV T KYEFMGKDKK MNQIFNKSMV DVCATEMKRM LEIYTGFEIG
STLVDVGGGS GRNLELIISK YPLIKGINFD LPQVIENAPP LSGIEHVGGD
MFASVPQGDA MILKAVCHNW SDEKCIEFLS NCHKALSPNG KVIIVEFILP
EEPNTSEESK LVSTLDNLMF ITVGGRETE KQYEKLSKLS GFSKFQVACR
AFNSLGVMF YK MsCOMT (*Medicago sativa*; P28002) (SEQ ID NO: 83)
MGSTGETQIT PTHISDEEAN LFAMQLASAS VLP MILKSAL ELDLLEIIAK
AGPGAQISPI EIASQLPTTN PDAPVMLDRM LRLACYIIL TCSVRTQQDG
KVQRLYGLAT VAKYLVKNE D GVSISALNLM NQDKVLMESW YHLKDAVLDG
GIPFNKAYGM TAFEYHGTD P RFNKVENKGM SDHSTITMKK ILETYTGFE G
LKSLVDVGGG TGAVINTIVS KYPTIKGINF DLPHVIEDAP SYPGVEHVGG
DMFVSIPKAD AVFMKWICH D WSDEHCLKFL KNCYEALPDN GKVIVAECIL
PVAPDSSLAT KGVVHIDVIM LAHNPGGKER TQKEFEDLAK GAGFQGFKVH
CNAFNTYIME FLKKV MtiOMT (*Medicago truncatula*; Q29U70) (SEQ ID NO:
84) MAFSTNGSEE SELYHAQIHL YKHVYNFVSS MALKSAMELG IADAIHNHGK
PMTLSELASS LKLHPSKVNI LHRFLRLLTH NGFFAKTIVK GKEGDEEEEI
AYSLTPPSKL LISGKPTCLS SIVKGALHPS SLDMWSSSKK WFNEDKEQTL
FECATGESFW DFLNKDSESS TLSMFQDAMA SDSRMFKLVL QENKRVFEGL
ESLVDVGGGT GGVTKLIHEI FPHLKCTVFD QPQVVGNLTG NENLNFVGGD
MFKSIPSADA VLLKWVLHDW NDEQSLKILK NSKEAISHKG KDGKVIIDI
SIDETSDDR G LTELQLDYDL VMLTMFLGKE RTKQEWELI YDAGFSSYKI
TPISGFKSLI EVYP MtiOMT3 (*Medicago truncatula*; Q06YR3) (SEQ ID NO: 85)
MASSINNRP SEIFKAQALL YKNMYAFVDS MSLKWSIEMN IPNIIHNHGK
PITLSNLVSI LQIPSTKVDN VQRLMRYLAH NGFFEITNQ ELENEEEAYA
LTVASELLVK GTCLCLAPMV ECVLDPTLST SFHNLKKWVY EEDLTLFAVN
LGCDLWEFLN KNPEYNTLYN DALASDSKMI NLAMKDCNLV FEGLESIVDV
GGNGTTGKI ICETFPKLT C VVFD RPKVVE NLCGSNNLT Y VGGDMFISVP
KADAVLLKAV LHDWTDKDCI KILKKCKEAV TSDGKRGKVI VIDMVINEKK
DENQLTQIKL LMNVTISCVN GKERN EEWK KLFIEAGFQD YKISPFTGLM
SLIEIYP ScNOMT (*Streptomyces carzinostaticus*; Q84HC8) (SEQ ID NO: 86)
MGKRAAHIGL RALADLATPM AVRVAATLRV ADHIAAGHRT AAEIASAAGA
HADSLDRLLR HLVA VGLFTR DGQGVYGLTE FGEQLRDDHA AGKRKWLDMN
SAVGRGDLGF VELAH SIRTG QPAYPVRYGT SFWEDLGSDP VLSASFDTLM
SHHLELDYTG IAAKYDWAAL GHVVDVGGGS GGLLSALLTA HEDLSGTVLD
LQGPASAAHR RFLDTGLSGR AQVVVGSFFD PLPAGAGGYV LSAVLHDWDD
LSAVAILRRC AEAAGSGGVV LVIEAVAGDE HAGTGMDLRM LTYFGGKERS
LAELGELAAQ AGLAVRAAHP ISYVSIVEMT AL LpCOMT (*Loium perenne*;
Q9ZTU2) (SEQ ID NO: 87) MGSTAADMAA SADEDACMFA LQLASSSVLP
MTLKNAIELG LLEILVAAGG KSLTPTEVAA KLPSAANPEA PDMVDRILRL
LASYNVVTCL VEEGKDGRLS RSYGAAPVCK FLTPNEDGVS MAALALMNQD
KVLME SWYYL KDAVLDGGIP FNKAYGMSAF EYHGTDPREN RVFNEG MKNH
SIITKKLLE LYHGFEG LGT LVDVGGGVGA TVAAIAAHYP TIKGVNFDLP
HVISEAPQFP GVTHVGGDMF KEVPSGDTIL MKWILHDWSD QHCATLLKNC
YDALPAHGKV VLVQCILPVN PEANPSSQGV FHVDMIMLAH NPGGRERYER
EFQALARGAG FTGVKSTYIY ANAWAIEFTK CHIEOMT (*Clarkia breweri*; 004385)
(SEQ ID NO: 88) MGSTGNAEIQ IIPTHSSDEE ANLFAMQLAS AAVLPMALKA

AIELDVLEIM AKSVAPVSL SPAEIAAQLP TTNPEAPVML DRVLEALASY
 SVVTTYTLREL PSGKVERLYG LAPVCKFLT K NEDGVSLAPF LLTATDKVLL
 EPWFYLKDAI LEGGIPFNKA YGMNEFDYHG TDHRFNKVFN KGMSSNSTIT
 MKKILEMYNG FEGLTTIVDV GGGTGAVASM IVAKYPSINA INFDLPHVIQ
 DAPAFSGVEH LGGDMFDGVP KGDAIFIKWI CHDWSDEHCL KLLKNCYAAL
 PDHGKVIVAE YILPPSPDPS IATKVVIHTD ALMLAYNPGG KERTEKEFQA
 LAMASGFRGF KVASCAFNTY VMEFLKTA PIOMT (*Planctopirus limnophila*;
 D5STZ7) (SEQ ID NO: 89) MAVKDALRFP PTDVTPIFDL FRGNFATELL
 AASVAHLHVF DILNESPLSL DELQRRVLVS ERATQVLVTG LCAMQLLTR
 LAGEIDLTP L ARNHLVTTSP FSVGGYISLA AQSAGTLALV ERLKSDRPEG
 AESEQGAafi FREGSESAMD REDSARFLTL SLAGRAWNVA PRFADVLPAg
 QPGKILKDNS GSSGRVLLDV AGGSGIYTMA VLQKYPTWRG IIFDRPEVLK
 IAAELAEQTG VRDRLELHAG DMWVDPFPPA DDILLSNVLH DWDRPQCARL
 VAKATSGLPE GGRLLIHDVL LNSDLTGPLE IALYSLALFS LTEGRAYSLE
 EYRGWIAGAD LKYVDCIPTS AHGHLILSEK VTINOMT (*Thalictrum flavum*;
 Q5C9L7) (SEQ ID NO: 90) MEMINKENLS SQAKLWNFIY GFADSLVLKS
 AVQLDLANII HNHGSPMTLS ELSLHLPSQP VNQDALYRVL RYLVHMKLFT
 KSSIDGELRY GLAPPAKFLV KGWDKCMLGA ILTITDKDFM APWHYLKEGI
 LNDGSTSTAF EKALGTNIWD YMAEHPEKNQ LFNEGMANDT RLIMSALVKE
 CSSMFDGITT IVDVGGGTGT AVRNIKAFP HIKCTVYDLP HVIADSPGYT
 EINSIQGDMF KYIPNADAIM MKCILHDWDD KECIEILKRC KDAVPRDGGK
 VIIIDIILDV KSEHPYTKMR LTLDLDMMLN TGGKERTEEE WKKLIHDAGY
 KGYKITHISA VQSVIEAYPY KpBOMT (*Kitagawia praeruptora*; A0A166U5H3)
 (SEQ ID NO: 91) MAGMKTSPSQ DEEACVLAIQ LATSTVLPMI LKSAIELDIL
 NTISKAGPGN YLSPSDLASK LLMSNPAPI MLERILRVLA TYKVLGCKPS
 ELSDGEVEWL YCWTPVCKFL SNNEGDASIA PLLLVHQDQV PMKSWYHLTD
 AILDGGTAFN KAYGMNIFY ASQDPQFNKV FNRSMAGHST ITMKKILETY
 NGFEGLSIV DVGGGSGATL NMISKYPTI KGINFDLPHV VGDSPHHPGV
 EHVGGDMFAS VPKGDAIFLK WIFHSWDED CLRILKNCYE ALADNKKVIV
 AEFIPEVPG GSDDATKSVV HLDVMLAYV PGGKERTEKE FEALATSAGF
 KSFRKVCCAF NTWIMEFSK PsSOMT (*Papaver somniferum*; I3V6A7) (SEQ ID
 NO: 92) MATNGEIFNT YGHNRTATV TKITASNESS NGVCYLSETA
 NLGKLICIPM ALRAAMELV FQLISKFGTD AKVSASEIAS KMPNAKNNPE
 AAMYLDRLR LLGASSILSV STTKKSINRG GDDVVVHEKL YGLTNSSCCL
 VPRQEDGVSL VEELLFTSDK VVVDSEFKLK CVVEEKDSVP FEVAHGAKIF
 EYAATEPRMN QVENDGMAVF SIVVFEAVFR FYDGFLDMKE LLDVGGGIGT
 SVSKIVAKYP LIRGVNFDLP HVISVAPQYP GVEHVAGDMF EEVPKGQNML
 LKWVLHDWGD ERCVKLLKNC WNSLPVGGKV LIIEFVLPNE LGNNAESFNA
 LIPDLLLMAL NPGGKERTIS EYDDLGAAG FIKTIPIPI NGLHVIEFHK FaOMT
 (*Fragaria ananassa*; Q9M602) (SEQ ID NO: 93) MGSTGETQMT PTHVSDEEAN
 LFAMQLASAS VLPMLVKAAL ELDLLEIMAK AGPGSFLSPS DLASQLPTKN
 PEAPVMLDRM LRLASYSIL TCSLRTLDPG KVERLYCLGP VCKFLTKNED
 GVSIAALCLM NQDKVLVESW YHLKDAVLDG GIPFNKAYGM TAFDYHGTDP
 RENKVENKGM ADHSTITMKK ILETYKGFEG LKSIVDVGGG TGAVVNMIVS
 KYPSIKGINF DLPHVIEDAP QYPGVQHVGG DMFVSVPKGN AIFMKWICH
 WSDEHCKIFL KNCYAALPDD GKVILAECIL PVAPDTSLAT KGVVHMDVIM
 LAHNPGGKER TEQEFEALAK GSGFQGIRVC CDAFNTYVIE FLKKI

[0236] It is understood that the disclosed method and compositions are not limited to the particular methodology, protocols, and reagents described as these can vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is

not intended to limit the scope of the present invention which will be limited only by the appended claims.

[0237] Disclosed are materials, compositions, and components that can be used for, can be used in conjunction with, can be used in preparation for, or are products of the disclosed method and compositions. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective combinations and permutation of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. For example, if an OMT is disclosed and discussed and a number of modifications that can be made to a number of enzymes including the OMT are discussed, each and every combination and permutation of OMT and the modifications that are possible are specifically contemplated unless specifically indicated to the contrary. Thus, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F and an example of a combination molecule, A-D is disclosed, then even if each is not individually recited, each is individually and collectively contemplated. Thus, in this example, each of the combinations A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and F; and the example combination A-D. Likewise, any subset or combination of these is also specifically contemplated and disclosed. Thus, for example, the sub-group of A-E, B-F, and C-E are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and F; and the example combination A-D. Further, each of the materials, compositions, components, etc. contemplated and disclosed as above can also be specifically and independently included or excluded from any group, subgroup, list, set, etc. of such materials. These concepts apply to all aspects of this application including, but not limited to, steps in methods of making and using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods, and that each such combination is specifically contemplated and should be considered disclosed.

[0238] It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise. Thus, for example, reference to “a OMT” includes a plurality of such OMTs, reference to “the OMT” is a reference to one or more OMTs and equivalents thereof known to those skilled in the art, and so forth.

[0239] Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude, for example, other additives, components, integers or steps.

[0240] “Optional” or “optionally” means that the subsequently described event, circumstance, or material may or may not occur or be present, and that the description includes instances where the event, circumstance, or material occurs or is present and instances where it does not occur or is not present.

[0241] Unless the context clearly indicates otherwise, use of the word “can” indicates an option or capability of the object or condition referred to. Generally, use of “can” in this way is meant to positively state the option or capability while also leaving open that the option or capability could be absent in other forms or embodiments of the object or condition referred to. Unless the context clearly indicates otherwise, use of the word “may” indicates an option or capability of the object or condition referred to. Generally, use of “may” in this way is meant to positively state the option or capability while also leaving open that the option or capability could be absent in other forms or embodiments of the object or condition referred to. Unless the context clearly indicates otherwise, use of “may” herein does not refer to an unknown or doubtful feature of an object or condition.

[0242] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, also specifically contemplated and

considered disclosed is the range from the one particular value and/or to the other particular value unless the context specifically indicates otherwise. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another, specifically contemplated embodiment that should be considered disclosed unless the context specifically indicates otherwise. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint unless the context specifically indicates otherwise. It should be understood that all of the individual values and sub-ranges of values contained within an explicitly disclosed range are also specifically contemplated and should be considered disclosed unless the context specifically indicates otherwise. Finally, it should be understood that all ranges refer both to the recited range as a range and as a collection of individual numbers from and including the first endpoint to and including the second endpoint. In the latter case, it should be understood that any of the individual numbers can be selected as one form of the quantity, value, or feature to which the range refers. In this way, a range describes a set of numbers or values from and including the first endpoint to and including the second endpoint from which a single member of the set (i.e. a single number) can be selected as the quantity, value, or feature to which the range refers. The foregoing applies regardless of whether in particular cases some or all of these embodiments are explicitly disclosed.

[0243] Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosed method and compositions belong. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present method and compositions, the particularly useful methods, devices, and materials are as described. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such disclosure by virtue of prior invention. No admission is made that any reference constitutes prior art. The discussion of references states what their authors assert, and applicants reserve the right to challenge the accuracy and pertinency of the cited documents. It will be clearly understood that, although a number of publications are referred to herein, such reference does not constitute an admission that any of these documents forms part of the common general knowledge in the art.

[0244] Although the description of materials, compositions, components, steps, techniques, etc. can include numerous options and alternatives, this should not be construed as, and is not an admission that, such options and alternatives are equivalent to each other or, in particular, are obvious alternatives. Thus, for example, a list of different OMTs does not indicate that the listed OMTs are obvious one to the other, nor is it an admission of equivalence or obviousness.

[0245] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the method and compositions described herein. Such equivalents are intended to be encompassed by the following claims.

Claims

1. A modified Type 1 O-methyltransferase (OMT), wherein the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid or wherein the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid.
2. The modified OMT of claim 1, wherein the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid or wherein the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.
3. The modified OMT of claim 1, wherein the substituting polar amino acid is independently Asn, Gln, Thr, or Ser.
4. The modified OMT of claim 1, wherein the substituting polar amino acid is Asn.
5. The modified OMT of claim 1, wherein the substituting hydrophobic amino acid is

independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala.

6. The modified OMT of claim 1, wherein the substituting hydrophobic amino acid is Phe.

7. The modified OMT of claim 1, wherein the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCaldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCaldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMB, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SICOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIEOMT, *Planctopirus limnophila* PlOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

8. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

9. The modified OMT of claim 1, wherein the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

10. The modified OMT of claim 1, wherein the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

11. The modified OMT of claim 1, wherein the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

12. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

13. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

14. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

15. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

16. The modified OMT of claim 1, wherein the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

17. A solution comprising the modified OMT of claim 1, wherein the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions.

- 18.** A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising bringing into contact, in a solution, the modified OMT of claim 1, a substrate for the modified OMT and S-adenosylmethionine; and incubating under suitable conditions to produce the O-methylated stilbenoid or hydroxycinnamic acid.
- 19.** A plant that expresses the modified OMT of claim 1.
- 20.** A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising growing the plant of claim 19 and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.
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